INVESTING & MARKETS (INDIA)

Analysing a Stock: Fundamental & Technical

Reading a listed Indian company the way an analyst does — the three statements line by line with a full worked set of accounts, every ratio that matters, relative and intrinsic valuation including a complete DCF and reverse DCF, moats and management quality, forensic red flags, sector-specific frameworks for banks and NBFCs, and technical analysis presented with the actual evidence rather than the folklore.

stock analysis · fundamental analysis · technical analysis · valuation · dcf · financial statements · ratio analysis · balance sheet · cash flow · india · annual report · ind as · moats · red flags · screening · investing

Study Pack · Comprehensive reference & interview prep · ~158 min read · 34,765 words

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What Analysis Is Actually For

Most people believe stock analysis is about predicting what a share price will do. It is not. Analysis is about estimating the range of cash a business can generate over its remaining life, assigning rough probabilities to the outcomes in that range, and then comparing that estimate against the price the market is asking. You will be wrong constantly. The discipline is in being wrong in ways that are survivable, and in knowing which of your assumptions broke.

❌ **This is financial education, not financial advice**
Nothing in this guide is a recommendation to buy or sell any security. Every company used for worked financial examples is **hypothetical and invented**. Where a real listed company is named, it is only as a neutral illustration of a structural point, never as a suggestion to invest. **Past performance does not predict future returns.** Equity investing puts your capital at risk and you can lose all of it — individual listed Indian companies have gone to zero, and broad Indian indices have fallen more than 50% from peak to trough within living memory. For advice specific to your circumstances, consult a **SEBI-registered Investment Adviser (RIA)** and verify their registration number on the SEBI intermediary portal before paying for anything.

⚠ **Every rate, rule, threshold and accounting standard: as at FY 2025-26 / late 2026**
Ind AS requirements, SEBI LODR disclosure obligations, tax rates, the corporate tax rate used in the worked examples, G-Sec yields, equity risk premium estimates and disclosure formats all change. **Verify before relying on any number here** at **sebi.gov.in**, **mca.gov.in**, **incometax.gov.in**, **rbi.org.in**, **nseindia.com** and **bseindia.com**. Treat every figure in this guide as illustrative of a method, not as a current quotation.

The only equation that matters

Everything in fundamental analysis — every ratio, every margin, every screening filter — exists to help you populate the terms of one equation.

TEXT

```
THE VALUE OF A BUSINESS

                 n
    Value  =    sum    [  CF(t)  /  (1 + r)^t  ]
                t=1

  CF(t) = cash available to owners in period t
  r     = the return you demand for bearing this risk
  n     = the life of the business

Three variables. That is the entire discipline.

WHAT EACH BRANCH OF ANALYSIS IS ACTUALLY DOING

  Reading the P&L           → estimating current CF and its
                              composition and durability
  Reading the balance sheet → checking whether the business
                              survives long enough to reach
                              year n, and how much capital it
                              consumes to grow CF
  Reading the cash flow     → verifying that the CF in the
                              P&L is real
  Ratio analysis            → measuring how much capital is
                              consumed per rupee of CF
  Moat analysis             → estimating how long CF can be
                              defended before competition
                              drives returns to the cost of
                              capital
  Valuation                 → converting all of the above
                              into a number, and comparing it
                              with the asking price
  Technical analysis        → says nothing about CF at all;
                              it is a study of the behaviour
                              of other buyers and sellers

Note the asymmetry: you can be approximately right about
CF and still lose money if you overpay. You can be wrong
about CF and still make money if you underpaid enough.
PRICE IS HALF THE ANALYSIS.
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You are estimating a distribution, not a point

The single most common beginner error is to produce one number — "this company will earn ₹450 crore next year, therefore the share is worth ₹820" — and then defend it. A professional estimate looks nothing like that. It looks like a distribution with explicit drivers.

Scenario	What has to be true	FY2030 revenue	FY2030 EBIT margin	Rough fair value / share	Su pr
Bear	A new entrant with cheaper imports takes 8% share; pricing power erodes; capex overruns	₹2,900 cr	11.0%	₹150–180	25
Base	Industry grows at nominal GDP + 3%; company holds share; margin creeps up with scale	₹3,864 cr	16.5%	₹240–290	5(
Bull	The export order book converts; the new plant runs at 85% utilisation two years early	₹4,600 cr	19.0%	₹380–430	2(
Disaster	Promoter pledge invoked, covenant breach, forced asset sale, or an accounting restatement	—	—	₹0–50	5%

Expected value here is roughly `0.25×165 + 0.50×265 + 0.20×405 + 0.05×25 = 41 + 133 + 81 + 1 = ₹256`. The number is less interesting than the shape. A 5% chance of near-total loss changes how much of your portfolio belongs in this position far more than the ₹256 does.

Why the "disaster" row exists on every thesis

Permanent capital loss and temporary price decline are entirely different risks and require entirely different responses. A 40% drawdown in a sound business with net cash is an opportunity. A 40% drawdown in a leveraged business whose lenders are getting nervous is the first instalment of a 95% loss. Section 11 of this guide is about telling the two apart *before* the drawdown.

Fundamental, technical, quantitative: what each actually claims

School	Core claim	Input data	Typical horizon	What the evidence supports
Fundamental	Price eventually converges on the present value of cash flows	Financial statements, filings, industry data, management	2–10 years	Value and quality characteristics have shown long-run premia in most markets, including India, with long and painful periods of underperformance
Technical	Price and volume history contain exploitable information about future price	Price, volume, derived indicators	Minutes to months	**Time-series momentum and trend** have survived out-of-sample testing across markets and decades. **Most named chart patterns have not.**
Quantitative / factor	Systematic exposure to statistical characteristics (value, momentum, quality, low volatility, size) earns a premium	Large cross-sectional datasets	Years	Factor premia are documented in Indian data too, but after costs, turnover and crowding, realised premia are materially smaller than backtests suggest
Narrative / tips	Someone knows something	Messaging groups, television	Days	Nothing. Acting on unpublished price-sensitive information is also an offence under

				SEBI (PIT) Regulations, 2015

What you are competing against

Before investing a single rupee of research effort, be clear about who is on the other side of your trade. In an Indian large-cap, your counterparty is frequently a well-staffed institutional desk with a market terminal, channel checks, paid expert networks, a direct line to the company's investor relations team, and twenty years in the sector. You will not out-research them on next quarter's revenue.

WHERE YOU CANNOT WIN

Short-horizon earnings prediction, speed of reaction to news, access to management, breadth of coverage, cost of execution. Accept this permanently.

WHERE YOU CAN WIN

Time horizon. A fund manager judged quarterly cannot hold a stock through three flat years. You can. This is the single largest structural edge available to an individual.

AND HERE

Opportunity set. You can own a ₹900 crore company; a ₹50,000 crore fund cannot take a meaningful position in one without moving the price. Analyst coverage thins dramatically below ₹5,000 crore market capitalisation.

AND HERE

No mandate. You may hold 60% cash for two years. No professional can. You are never forced to be invested.

AND HERE

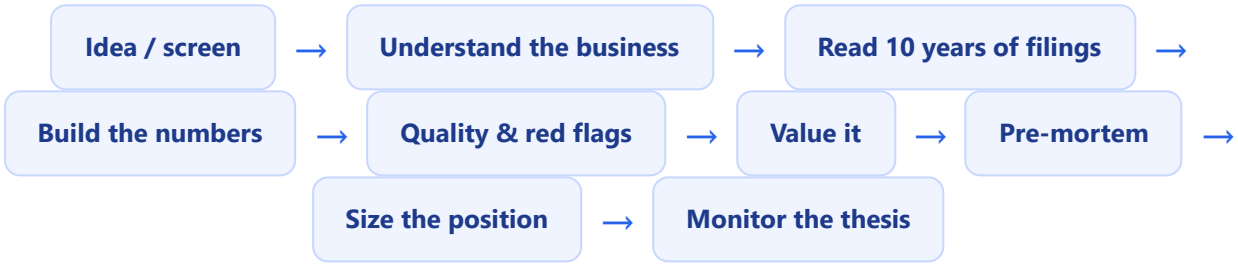
No career risk. Being unconventional and wrong ends careers. It does not end yours. This permits genuinely contrarian positions.

THE CATCH

Every one of these edges is *behavioural*. They only exist if you actually use them. Many individual investors trade more frequently than professionals, which converts every structural advantage into a disadvantage.

3	2	10	1
STATEMENTS TO READ	VARIABLES MOVE PRICE	YEARS OF DATA, MINIMUM	REAL EDGE: HORIZON

The analysis workflow, end to end



Note what is absent: there is no step called "check the chart to see if it is going up". Technical inputs, if you use them at all, belong at the sizing and timing stage, after the business question is settled — and sections 14 and 15 are blunt about how much weight they deserve.

How long does one company actually take?

Where the Information Lives

Every fact you need about a listed Indian company is published, free, in a handful of documents, most of which almost nobody reads. The competitive advantage available to a patient individual is not secret information — it is public information that institutional analysts skim and retail investors skip entirely. This section is a map of the corpus.

The primary sources, ranked by information density

#	Document	Frequency	Where to get it	Why it matters	Pa
1	**Annual Report**	Yearly, roughly three months after 31 March	Company website "Investors" section; exchange filings; MCA	The only document containing audited accounts with full notes, the auditor's report, related-party disclosures and the directors' report. Everything else is derivative.	15
2	**Concall transcript**	Quarterly	Company website; exchange filings (SEBI LODR requires disclosure)	The only place where management answers questions it did not write. Analyst scepticism is visible. Evasions are visible.	15
3	**Quarterly results filing**	Quarterly, within the SEBI-prescribed window after quarter end	NSE/BSE announcements; company site	Limited-review (not fully audited) P&L, segment data, and a half-yearly balance sheet and cash flow statement under LODR Reg. 33	4-
4	**Investor presentation**	Quarterly, and on events	Company website; exchange filings	Management's chosen framing. Useful for operating metrics	2(

				(volumes, utilisation, order book) that are not in the statutory accounts — and for spotting which metric they quietly stopped reporting.	
5	**Credit rating rationale**	At least annually	CRISIL, ICRA, CARE, India Ratings, Acuité websites — free	An independent third party with access to management and unpublished data, writing about leverage, liquidity and covenants. Chronically under-read by equity investors.	4-
6	**Shareholding pattern**	Quarterly, within the prescribed days of quarter end	NSE/BSE	Promoter holding, **pledged shares**, FII/DII/mutual fund holding, public float. The pledge column is often the single highest-value number on the page.	3-
7	**DRHP / RHP** (for IPOs)	Once, pre-listing	SEBI website; merchant banker sites	The most detailed document any Indian company ever publishes about itself: industry data, competitive	4(

				positioning, risk factors, litigation, promoter history, use of proceeds. Worth reading for the *competitors* of a listed company you follow.	
8	**Exchange announcements**	Continuous	NSE/BSE corporate announcements	Material events under LODR Reg. 30: order wins, resignations, auditor changes, litigation, plant shutdowns, board meetings. Set up alerts.	1-
9	**AGM**	Yearly	Attend, or join the webcast; e-voting is mandatory for listed companies	You may ask a question as a shareholder. The quality of the answers tells you a great deal about the board's attitude to minority holders.	—
10	**Sector regulators and associations**	Varies	SIAM, IBA, RBI bulletins, TRAI, PNGRB, CEA, IBEF, industry bodies	Independent volume and pricing data with which to triangulate a company's claims about market share and industry growth	—

How to read an annual report in the right order

Reading a 300-page annual report front to back is the slowest possible route. The first sixty pages are marketing. Read in this order instead.

1. **Auditor's Report**
   Start at the back. Look for the opinion type (unmodified, qualified, adverse or disclaimer), the **Key Audit Matters** — these are the things the auditor found hardest, which is a free list of where the accounting risk sits — and the **CARO 2020** annexure, which asks pointed questions about defaults, undisclosed income, fund diversion and whistle-blower complaints.

2. **Notes: related-party transactions**
   Who is the company trading with, and are any of them owned by the promoter family? Quantify as a percentage of revenue and of purchases. A rising trend here is one of the strongest single red flags in Indian small- and mid-caps.

3. **Notes: contingent liabilities and commitments**
   Tax disputes, guarantees given for subsidiaries or group companies, pending litigation, capital commitments. These are real obligations that never appear on the face of the balance sheet.

4. **Cash flow statement**
   Before the P&L. Ten years of CFO compared against ten years of PAT answers the most important question about the accounts in one minute.

5. **Balance sheet and P&L with their notes**
   Never read a line item without its note. "Other expenses" of ₹456 crore is meaningless; the note breaking it into power and fuel, freight, job work, CSR, professional fees and "miscellaneous" is not.

6. **Management Discussion and Analysis**
   Mandatory under LODR Schedule V. Mediocre companies fill it with industry boilerplate copied from a consultancy report. Good ones explain variances. Read last year's MD&A and check what was promised against what happened.

7. **Directors' Report and corporate governance report**
   Board composition, independent directors and their other directorships, attendance, audit committee, remuneration, and any resignation together with its stated reason.

8. **Segment reporting note**

Where revenue and — crucially — *segment results and capital employed* actually sit. A company with one profitable segment subsidising two loss-making ones looks entirely different once disaggregated.

9

Chairman's letter, finally

Read it last, after you know the numbers, so that you can judge the honesty of the framing rather than be anchored by it. The test: does it discuss what went wrong, in specifics, with numbers?

✦ **Read ten years at once, not one**

A single annual report tells you almost nothing. Download ten, open them side by side, and extract one line of data per year into a spreadsheet. Trends, inflections and quietly abandoned promises are only visible in the series. The mechanical act of typing ten years of numbers by hand — rather than importing them — is also the fastest way to learn what the line items actually mean.

How to listen to an earnings call

The transcript is more useful than the audio, because you can search it. A concall has a predictable structure and the information is unevenly distributed across it.

Segment	Typical length	Information value	What to extract
Safe-harbour statement	1 min	Zero	Skip
Management opening remarks	10–15 min	Low to moderate	Scripted. Useful mainly for the specific guidance numbers given — capex plan, margin target, volume outlook — which you will hold them to next quarter.
Analyst Q&A	35–50 min	**High**	This is the document. Which analysts ask hostile questions, what they are worried about, and whether management answers or deflects.
Closing remarks	2 min	Zero	Skip

THE SPECIFIC THINGS TO LOOK FOR IN THE Q&A

- ☐ **Repeated questions.** If three different analysts circle the same issue, management has not answered it. Note the issue.
- ☐ **Answer substitution.** Question: "Why did gross margin fall 180 basis points?" Answer: "We are very confident about the demand environment." That is a non-answer, and it is a data point.
- ☐ **Guidance drift.** Track one number — say FY2027 capex — across eight consecutive calls. "₹400 crore" becoming "₹400–450 crore" becoming "around ₹500 crore, with some spillover into FY2028" is a cost overrun being disclosed in slow motion.
- ☐ **Metric abandonment.** A company that reported capacity utilisation for twelve quarters and then stopped is telling you that utilisation fell.
- ☐ **Who is on the call.** If the CFO stops attending, find out why. If the same two analysts from boutique firms are the only participants, institutional interest has gone.
- ☐ **Tone towards questions about cash.** Defensive responses to straightforward working-capital questions correlate with problems more reliably than almost anything else on a call.
- ☐ **Promises with dates.** Write them into a file with the date. Review in four quarters. A management team's hit rate on its own stated timelines is a direct measure of both competence and candour.

> ℹ **A free and underused resource: your competitors' calls**
>
> If you follow a cement company, read the concalls of four other cement companies. Industry pricing, regional demand, freight costs and capacity additions get discussed from five different angles, and inconsistencies between what the companies claim about market share are very informative. Most retail investors read only the filings of the company they own — which is exactly the wrong way round.

Data sources and their failure modes

Source type	Good for	Specific risks
Exchange websites (NSE, BSE)	Primary filings, shareholding, announcements, bulk and block deals, corporate actions — all free and authoritative	Poor search; the data is raw. This is the source of truth; everything else is a copy of it.
Screeners and aggregators	Fast ten-year summaries, ratio computation, screening	**Standardisation errors.** Aggregators map Indian company line items into their own schema and get it wrong, especially for finance companies, holding companies and anything with large other income or exceptional items. **Always verify a ratio against the annual report before acting on it.** Consolidated-versus-standalone confusion is endemic.
Broker research	Industry structure, channel data, model templates	Structural optimism: sell ratings are a small minority of published recommendations across the industry, because they damage banking and corporate-access relationships. Read the assumptions; ignore the target price.
Business media	Event awareness	Almost no information content about value. Headlines are a sentiment indicator, not an input.
Social media and messaging groups	Occasionally, genuinely good independent analysis	Pump-and-dump schemes in micro-caps are a live SEBI enforcement priority. Anyone giving stock-specific advice for a fee without an RIA registration number is operating illegally. Check the number on the SEBI portal.
Company investor relations	Clarifying a fact that has already been disclosed	They cannot give you unpublished price-sensitive information, and you must not ask for it — receiving and trading on it is an offence under SEBI (PIT) Regulations, 2015 regardless of whether you solicited it.

⚠ **Consolidated, not standalone**

For any company with subsidiaries, joint ventures or associates, the **consolidated** accounts describe the economic entity you own a share of. Standalone accounts exclude subsidiaries and can present a radically flattering picture — a parent whose standalone books look clean while debt and losses sit in step-down subsidiaries is a recurring pattern in Indian corporate distress. Use standalone only to isolate the parent's own operations as a diagnostic, never as your primary view. Many screeners default to standalone.

Building your own data file

Before any analysis, build a ten-year sheet by hand. This is tedious and it is the highest-return few hours in the entire process.

TEXT

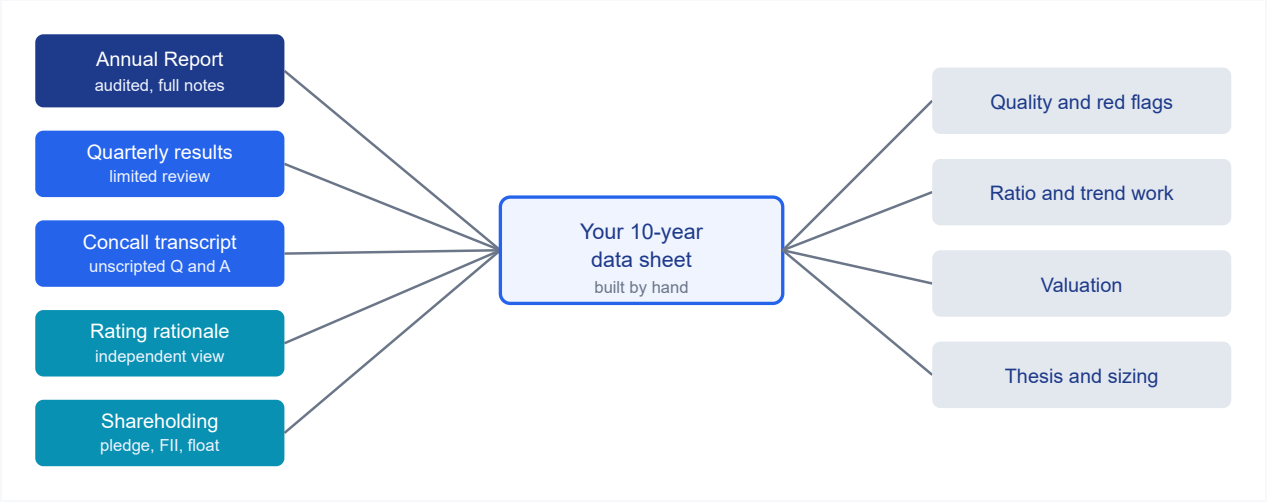
MINIMUM 10-YEAR DATA SHEET (consolidated, Rs crore)

P&L	BALANCE SHEET
Revenue	Total equity
Gross profit	Gross debt
EBITDA	Cash + liquid investments
Depreciation	Net debt
Finance cost	Net block (PPE)
Other income	CWIP
Exceptional items	Goodwill + intangibles
PBT	Inventories
Tax	Trade receivables
PAT	Trade payables
Minority interest	Total assets
EPS (diluted)	Capital employed

CASH FLOW	OPERATING / OTHER
CFO	Shares outstanding (diluted)
Capex	Promoter holding %
CFO − capex (FCF)	Promoter pledge %
Dividend paid	Capacity / volume (if given)
Buyback	Employee count
Equity raised	Auditor name
Net debt change	Effective tax rate

DERIVED (compute these, do not copy them)	
Revenue growth %	ROE %
Gross margin %	ROCE %
EBITDA margin %	ROIC %
PAT margin %	CFO / EBITDA %
Inventory days	Net debt / EBITDA
Receivable days	Interest coverage
Payable days	Capex / depreciation
Cash conversion cycle	Dividend payout %

RULE: compute every derived number yourself from the raw ones. If your ROCE disagrees with the screener's, find out why BEFORE you trust either. The disagreement is almost always in the denominator.



The information funnel: five free primary sources collapse into one hand-built data sheet, which feeds every downstream judgement.

What a DRHP gives you that nothing else does

Set up a monitoring routine before you need it

For each company you follow: subscribe to the exchange announcement alerts, diarise the results date, and keep one running text file per company with dated entries. Every time you learn something — a guidance number, an analyst's concern, a competitor's capacity announcement — add a dated line. After two years this file is worth more than any report you could buy, because it is a record of what you believed and when, which is the only way to find out whether your judgement is improving.

The Profit & Loss Statement, Line by Line

The P&L answers one question: over the past twelve months, did the company sell things for more than it cost to make and sell them? It is the most-read and least-reliable of the three statements, because almost every line involves a judgement call by management, and judgement calls are where manipulation lives. Read it with its notes, and read ten years of it at once.

Meet Bharat Widgets Ltd

Everything that follows uses one invented company, **Bharat Widgets Ltd (BWL)**, a mid-sized engineering components manufacturer with 12 crore shares outstanding at a face value of ₹10. Its full accounts are carried consistently through the P&L, balance sheet, cash flow, ratio, and DCF sections, so that every number ties back. BWL does not exist. No figure here describes any real company.

The statutory format under Ind AS / Schedule III

Indian listed companies present the P&L in a format prescribed by Schedule III to the Companies Act, 2013, as modified for Ind AS. The order is fixed, which makes comparison across companies straightforward once you learn it.

TEXT

BHARAT WIDGETS LTD — CONSOLIDATED STATEMENT OF PROFIT AND LOSS FOR THE YEAR ENDED 31 MARCH 2025
(Rs crore, except per-share data)

		FY2025	FY2024
I.	REVENUE		
	Revenue from operations	2,400	2,120
	Other income	36	31
	Total income	2,436	2,151
II.	EXPENSES		
	Cost of materials consumed	1,104	996
	Purchases of stock-in-trade	72	66
	Changes in inventories of finished goods and WIP	(48)	(22)
	Employee benefits expense	312	284
	Finance costs	58	61
	Depreciation and amortisation	132	118
	Other expenses	456	406
	Total expenses	2,086	1,909
III.	Profit before exceptional items and tax	350	242
IV.	Exceptional items (loss)	(24)	—
V.	Profit before tax	326	242
VI.	TAX EXPENSE		
	Current tax	92	66
	Deferred tax	(6)	(4)
	Total tax expense	86	62
VII.	PROFIT FOR THE YEAR	240	180
VIII.	OTHER COMPREHENSIVE INCOME		
	Items not reclassified to P&L		
	Remeasurement of defined benefit plans	(3)	2
	Tax on above	1	(1)
	Total OCI	(2)	1
	TOTAL COMPREHENSIVE INCOME	238	181
IX.	EARNINGS PER SHARE (face value Rs 10)		
	Basic	20.00	15.00
	Diluted	19.84	14.92

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Shares outstanding: 12.00 crore
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Every line, explained

Line	What it is	Judgement involved	What to check
Revenue from operations	Sales of goods and services, net of GST, trade discounts and returns. Recognised under Ind AS 115 when control passes to the customer.	**High.** Timing of control transfer, variable consideration, bill-and-hold arrangements, percentage-of-completion for long contracts, gross versus net (principal versus agent) presentation	Growth versus industry volume growth. Revenue growing far faster than the sector means share gain, a price increase, an acquisition, or a recognition change. Find out which.
Other income	Interest on deposits, dividend income, gains on investments, foreign-exchange gains, government grants, sale of scrap, write-backs of old provisions	**High.** This is where non-recurring items hide	As a percentage of PBT. If other income is 36 on PBT of 326, that is 11% — acceptable. If it is 40% of PBT, the "profit" is not from the business. Read the note; demand to know what it consists of.
Cost of materials consumed	Opening stock of raw materials + purchases − closing stock. The direct input cost.	Moderate. Inventory valuation method, provision for obsolescence	As a percentage of revenue, over ten years. This is the cleanest read on pricing power: a company that can pass on input-cost inflation holds this ratio steady through a commodity cycle.
Purchases of stock-in-trade	Finished goods bought for resale without further processing (a trading activity)	Low	A rising share of this line relative to manufactured sales means the company is becoming a trader — typically a lower-margin, lower-moat activity.
Changes in inventories of FG and WIP	An adjustment, not a cash cost. A *negative* figure (shown in brackets) means	Moderate	**This is the line beginners misread most often.** Production into inventory flatters current profit.

	inventory **increased** — the company produced more than it sold, so those costs are deferred to the balance sheet and profit is higher.		Cross-check against the inventory balance on the balance sheet and against inventory days.
Employee benefits expense	Salaries, bonuses, contributions to provident and gratuity funds, share-based payment expense, staff welfare	Moderate. Actuarial assumptions for gratuity; ESOP valuation	Revenue per employee and employee cost per employee, trended. For services businesses this is the dominant cost and the main operating-leverage lever.
Finance costs	Interest on borrowings, interest on lease liabilities under Ind AS 116, other borrowing costs, amortisation of transaction costs. **Excludes** interest capitalised into assets under construction.	Moderate. The capitalisation decision	Divide by average gross debt to get the implied interest rate. If that comes to 4% while bank lending rates are 9%, interest is being capitalised into CWIP — find the amount in the PPE note.
Depreciation and amortisation	The systematic write-off of PPE, right-of-use assets and intangibles over their useful lives	**High.** Useful life and residual value are management estimates; a change extends or compresses the charge	Compare depreciation to gross block (the implied rate) across ten years. A sudden fall in the rate usually means useful lives were extended — which raises reported profit with no change in the business. The change must be disclosed.

Other expenses	Everything else: power and fuel, freight and forwarding, job work, repairs, rent, rates and taxes, insurance, advertising, travel, legal and professional, CSR, bad debts written off, and "miscellaneous"	Moderate	Always read the note. A large or fast-growing "miscellaneous" bucket is a sign of either sloppiness or something being concealed. Check CSR against the Section 135 requirement; check whether bad debts written off are rising.
Exceptional items	Material, non-recurring items presented separately: impairments, restructuring, plant closure, litigation settlements, gains on asset sales	**Very high.** What counts as exceptional is a presentation choice	**Count them over ten years.** If a company reports "exceptional" items in seven of ten years, they are not exceptional — they are the business, and they belong in your normalised earnings. Also note the asymmetry: losses are exceptional, gains are "other income".
Current tax / deferred tax	Tax payable on this year's taxable income, and the change in timing differences between book and tax profit	Moderate	Effective tax rate = total tax ÷ PBT. BWL: 86 ÷ 326 = **26.4%**. If the effective rate is far below the statutory rate for several years, find the reconciliation note — tax holidays, SEZ benefits and brought-forward losses all expire.
Other comprehensive income	Gains and losses routed around the P&L: actuarial remeasurements, fair-value changes on certain investments, foreign operation translation, cash-flow hedge reserves	Moderate	Usually small, occasionally not. Large recurring OCI losses with clean reported profit deserve investigation — the economic loss is real, it is simply not in PAT.

Diluted EPS	PAT attributable to equity holders ÷ weighted average shares, adjusted for all potential dilution from ESOPs, warrants and convertibles	Low	**Always use diluted, never basic.** BWL's 0.8% dilution gap is modest. A 15% gap means a large option overhang that will transfer value from you to employees.

Building the margin ladder

The statutory format does not show gross profit, EBITDA or EBIT. You compute them. There is no single official definition of any of the three in Indian GAAP, which is precisely why you must compute them yourself and state your definition.

TEXT

```
BWL MARGIN LADDER - FY2025 (Rs crore)

REVENUE FROM OPERATIONS                          2,400   100.0%

Less: Cost of materials consumed       1,104
      Purchases of stock-in-trade          72
      Changes in inventories (FG/WIP)     (48)
      ---------------------------------------
      Total direct material cost        1,128    47.0%
                                        -----
GROSS PROFIT                                      1,272    53.0%

Less: Employee benefits expense           312    13.0%
      Other expenses                      456    19.0%
                                        -----
EBITDA                                              504    21.0%
  (check: 504 = PBT before exceptional 350
          + finance cost 58 + D&A 132
          - other income 36)

Less: Depreciation and amortisation       132     5.5%
                                        -----
EBIT (operating profit)                             372    15.5%

Add:  Other income                         36
Less: Finance costs                        58
                                        -----
PBT BEFORE EXCEPTIONAL ITEMS                        350    14.6%

Less: Exceptional items                    24
                                        -----
PROFIT BEFORE TAX                                   326    13.6%

Less: Total tax expense                    86
                                        -----
PROFIT AFTER TAX                                    240    10.0%

NORMALISED PAT (adding back the after-tax
exceptional loss at the 25.17% statutory rate):
      24 x (1 - 0.2517) = 17.96, say 18
      240 + 18 = 258  ->  normalised margin 10.75%
      Normalised diluted EPS = 258 / 12.1 = Rs 21.32
```

⚠ **There is no standard definition of EBITDA**

Some companies and most screeners include other income in EBITDA. BWL's EBITDA is ₹504 crore excluding other income and ₹540 crore including it — a 7% difference that flows straight into every EV/EBITDA multiple you compute. Since Ind AS 116 took effect, operating lease costs sit below EBITDA as depreciation on right-of-use assets plus interest on lease liabilities, so EBITDA for lease-heavy businesses (retail, aviation, hospitals) jumped overnight without anything changing economically. **State your definition, apply it to every peer identically, and never compare an EBITDA you computed with one you did not.**

What each margin actually tells you

Margin	BWL FY2025	Primarily driven by	What a sustained change means	Typical Indian range
Gross margin	53.0%	Product mix, pricing power, input costs, manufacturing yield	The single best proxy for competitive position. Falling gross margin while revenue grows almost always means the company is buying growth with price cuts.	Commodities 10–20%; auto components 25–40%; branded FMCG 45–60%; software products 70–90%
EBITDA margin	21.0%	Gross margin plus operating leverage on fixed costs	Expanding faster than gross margin means fixed costs are being spread over more volume — genuine operating leverage. Contracting while gross margin holds means cost indiscipline.	IT services 20–27%; cement 15–25%; pharma 18–25%; retail 6–10%; capital goods 10–15%
EBIT margin	15.5%	EBITDA less the real cost of the asset base	The most honest operating-profit measure, because depreciation is a genuine economic cost. Capital-intensive businesses with impressive EBITDA and thin EBIT are consuming their asset base.	Wide. Compare only within a sector.

PBT margin	13.6%	EBIT, other income, interest	The gap between EBIT and PBT margin is the cost of the capital structure. A wide gap means leverage is eating the operating result.	—
PAT margin	10.0%	Everything, including tax regime and one-offs	The most quoted and least informative margin. It mixes operations, financing, one-offs and tax policy into one number.	—

Ten-year margin trend: the actual analysis

One year of margins is a snapshot. The trend is the analysis. Here is BWL's history as you would lay it out.

FY	Revenue (₹ cr)	Growth	Gross %	EBITDA %	E
2016	1,012	—	50.1%	16.8%	1
2017	1,118	10.5%	50.4%	17.2%	1
2018	1,266	13.2%	51.0%	18.0%	1
2019	1,402	10.7%	51.3%	18.4%	1
2020	1,368	−2.4%	50.2%	16.6%	1
2021	1,490	8.9%	52.0%	18.9%	1
2022	1,742	16.9%	51.6%	19.4%	1
2023	1,916	10.0%	52.1%	19.8%	1
2024	2,120	10.6%	52.5%	20.3%	1
2025	2,400	13.2%	53.0%	21.0%	1

WHAT THIS TABLE SAYS

- **Gross margin is stable to slowly rising across a decade.** This is the headline finding. It survived FY2020. A company that holds gross margin through a demand shock has pricing power that customers accept.
- **EBITDA margin rose faster than gross margin** (+4.2 points versus +2.9 points). The extra 1.3 points is operating leverage: revenue grew 137% while employee and other costs grew less than proportionally.
- **EBIT margin rose faster still** (+4.1 points), which means depreciation grew more slowly than revenue — the existing asset base was sweated harder. This is good until it is not: at some point the plant is full and a large capex cycle begins. Check capacity utilisation disclosures.
- **Other income is consistently 8–14% of PBT.** Material but not dominant, and stable, so the profit is genuinely operating profit.
- **Three "exceptional" items in ten years.** That is borderline. Normalised ten-year PAT should add them back; if they recur again in FY2026, stop treating them as exceptional.

- **FY2020 was a real stress test.** Revenue fell 2.4% and EBITDA margin fell 180 basis points. That is a mild, well-managed downturn, and it is the most useful single year in the table for estimating the bear case.

✦ **The two-line test that catches most problems**
Put revenue growth and gross-margin change on the same chart for ten years. Three patterns matter. **Revenue up, gross margin up** — genuine competitive strength. **Revenue up, gross margin down sharply** — growth is being purchased with discounts and is likely worthless. **Revenue flat, gross margin up** — either a genuine mix improvement towards premium products, or the company has stopped chasing unprofitable volume, which is frequently a sign of good management. Add a third line for receivable days and you have caught most revenue-quality problems before opening a single note.

The P&L lines that get manipulated

Ranked roughly by how often the technique appears in Indian corporate accounting failures, and how hard it is to detect from outside.

Technique	How it works	Effect on P&L	Detection from published data
Channel stuffing	Push product to distributors near quarter end on extended credit, recognise the sale now	Revenue and profit pulled forward	Receivable days jump. Q4 revenue is a disproportionate share of the year. Q1 of the next year is unusually weak. Sales returns rise. **Easy to spot across four quarters.**
Capitalising operating expenses	Treat routine repairs, R&D, software or interest as an asset rather than an expense	Current profit up, assets up, future depreciation up	CWIP balance rising with no corresponding capacity increase. Capex far exceeding what the capacity addition would plausibly cost. Intangibles under development growing steadily. **Moderate difficulty.**
Extending depreciation lives	Decide that a machine lasts 20 years instead of 12	Depreciation falls, profit rises, no cash effect	Depreciation ÷ gross block falls with no change in asset mix. The change must be disclosed in the notes as a change in accounting estimate — read them. **Easy if you read the notes.**
Under-provisioning	Keep the provision for doubtful debts, inventory obsolescence or warranty artificially low	Profit up, assets overstated	Provision as a percentage of receivables falling while receivable ageing worsens. The ageing table is in the notes under Schedule III. **Moderate.**
Bill-and-hold and premature recognition	Recognise revenue before control actually transfers to the customer	Revenue pulled forward	Unbilled revenue or contract assets growing much faster than revenue. Common in EPC and project businesses. **Hard.**

Round-tripping through related parties	Sell to an entity controlled by the promoter; it "sells on" or simply holds the goods	Fictitious revenue and profit	Related-party revenue rising as a share of total. Receivables from related parties that never settle. **Hard, but the related-party note is the place to look.**
Classifying recurring costs as exceptional	Restructure every year, call it exceptional every year	"Adjusted" profit permanently overstated	Count exceptional items over ten years. If more than three, normalise them into the operating result. **Easy.**
Gross versus net revenue presentation	An agent reports the whole transaction value as its own revenue instead of just its commission	Revenue inflated; profit unchanged, margin optically collapsed	Implausibly thin margins on very large revenue. The Ind AS 115 principal-versus-agent note. **Easy once you look for it.**

✕ **The single strongest P&L red flag**

Reported profit that grows for several consecutive years while cumulative cash flow from operations does not. Every technique in the table above inflates PAT without generating cash. This is why the cash flow statement (section 5) is read *before* the P&L, and why the cumulative CFO-to-cumulative-PAT ratio over five to ten years is one of the highest-value numbers you can compute. BWL's five-year cumulative CFO of ₹1,318 crore against cumulative PAT of ₹922 crore — a ratio of 1.43 — is what a clean set of accounts looks like.

Quarterly results: what is and is not in them

The Balance Sheet, Line by Line

If the P&L is a film of the year, the balance sheet is a photograph taken at midnight on 31 March. It shows what the company owns, what it owes, and what is left over for shareholders. It is where solvency lives, where growth is funded, and where the evidence of P&L manipulation accumulates — because a fictitious profit has to be parked somewhere, and it is almost always parked in receivables, inventory or an intangible asset.

TEXT

BHARAT WIDGETS LTD – CONSOLIDATED BALANCE SHEET
AS AT 31 MARCH 2025 (Rs crore)

ASSETS	FY2025	FY2024
(1) NON-CURRENT ASSETS		
Property, plant and equipment	1,020	924
Capital work-in-progress	144	86
Right-of-use assets	108	114
Goodwill	180	180
Other intangible assets	96	102
Investments (non-current)	60	48
Other non-current assets	42	38
Total non-current assets	1,650	1,492
(2) CURRENT ASSETS		
Inventories	384	360
Trade receivables	432	396
Cash and cash equivalents	96	116
Other bank balances	54	48
Other financial assets	36	30
Other current assets	78	70
Total current assets	1,080	1,020
TOTAL ASSETS	2,730	2,512
EQUITY AND LIABILITIES		
(1) EQUITY		
Equity share capital	120	120
Other equity	1,380	1,200
Total equity	1,500	1,320
(2) NON-CURRENT LIABILITIES		
Borrowings	420	438
Lease liabilities	96	102
Deferred tax liabilities (net)	54	58
Long-term provisions	30	26
Total non-current liabilities	600	624
(3) CURRENT LIABILITIES		
Borrowings	180	174
Lease liabilities	24	22
Trade payables	288	252
Other financial liabilities	66	62
Other current liabilities	42	38

```
     Short-term provisions                18        14
     Current tax liabilities (net)        12         6
     ---------------------------------------------------
     Total current liabilities           630       568

     TOTAL EQUITY AND LIABILITIES      2,730     2,512

CHECK: 1,500 + 600 + 630 = 2,730 = total assets. OK.

DERIVED FIGURES
  Shares outstanding              12.00 crore
  Book value per share            1,500 / 12 = Rs 125.00
  Gross debt (ex-leases)          420 + 180   = Rs 600 cr
  Gross debt (incl. leases)       600 + 120   = Rs 720 cr
  Cash + bank balances            96 + 54     = Rs 150 cr
  Net debt (ex-leases)            600 - 150   = Rs 450 cr
  Net debt (incl. leases)         720 - 150   = Rs 570 cr
  Capital employed                2,730 - 630 = Rs 2,100 cr
    (also: equity 1,500 + debt 600 = 2,100)
  Net working capital             1,080 - 630 = Rs 450 cr
```

The asset side, line by line

Line	What it is	What to check	Red flag
Property, plant and equipment	Land, buildings, plant, machinery and vehicles at cost less accumulated depreciation ("net block"). The gross block, additions, disposals and accumulated depreciation are all in the PPE note.	Net block ÷ gross block tells you how old the asset base is. A ratio near 0.3 means heavily depreciated plant and an imminent replacement cycle. Compare revenue to net block for asset turnover.	Large revaluations. Assets held but idle. Land valued at a figure wildly different from its use.
Capital work-in-progress	Projects under construction, not yet ready for use, and therefore **not being depreciated**	How long has it sat there? Schedule III requires a CWIP ageing schedule and a disclosure of projects overdue or over budget. **Read it.**	**CWIP that never converts to PPE.** This is one of the most reliable indicators of either a stalled project or capitalised expenses parked out of sight. Depreciation that never starts flatters profit indefinitely.
Right-of-use assets	Leased assets brought on balance sheet under Ind AS 116, with a matching lease liability	Must be added to the asset base for any honest ROCE calculation, and the liability added to debt	Not a flag in itself, but ignoring it makes lease-heavy companies look far more capital-efficient than they are
Goodwill	The excess paid over the fair value of net assets in an acquisition. **Not amortised** under Ind AS; tested annually for impairment.	Goodwill as a percentage of total assets and of equity. BWL: 180 ÷ 1,500 = 12% of equity. Acceptable. Read the impairment-testing assumptions in the notes.	**Goodwill exceeding 30–40% of equity** means the company's book value is largely a record of prices paid for other companies. An impairment can erase a third of equity overnight.
Other intangible assets	Software, patents, trademarks, customer	Split between acquired and internally	**Large internally generated intangibles.**

	relationships, brands, and — critically — internally generated development costs capitalised under Ind AS 38	generated. Amortisation policy. Movement in "intangibles under development".	This is R&D that was expensed by a conservative company and capitalised by an aggressive one. It raises both profit and assets.
Investments (non-current)	Stakes in associates, joint ventures, unlisted entities, long-dated bonds and mutual funds	Who are these entities? Are any related to the promoter? Are they generating a return?	Investments in unlisted group entities that produce no income. A classic route for funds to leave the listed company.
Inventories	Raw materials, work in progress, finished goods, stores and spares, at the lower of cost and net realisable value	Inventory days. The split across the three categories. The provision for obsolescence.	**Inventory growing faster than revenue for several years.** Either demand has disappeared, or costs are being parked in inventory to avoid hitting the P&L. Watch finished goods specifically.
Trade receivables	Money owed by customers for goods already delivered	Receivable days. The **ageing schedule** required by Schedule III, split into under six months, six months to one year, one to two years, two to three years, and over three years. Provision as a percentage of the total.	**Receivables over one year that keep growing.** These are very unlikely to be collected and are usually still carried at full value. Also check receivables from related parties separately.
Cash and cash equivalents	Cash, current account balances, deposits with an original maturity of three months or less	Is it genuinely free? Check the note for balances held as margin money or pledged against borrowings.	**Large cash alongside large debt.** Sometimes legitimate (ring-fenced project debt, foreign subsidiary cash). Often not: if a company holds ₹800 crore of cash earning 6% while paying 9% on ₹800 crore of debt, it is destroying roughly ₹24 crore a year, and the question is whether that

			cash actually exists and is available.
Other bank balances	Deposits with maturity over three months, unpaid dividend accounts, margin money	Add to cash for net-debt purposes only where genuinely unencumbered	Rising restricted balances indicate lenders demanding more security
Other current / non-current assets	Advances to suppliers, prepaid expenses, balances with GST and customs authorities, capital advances, security deposits	Read the note. Large "advances to suppliers" or "capital advances" that persist for years are worth questioning.	Long-outstanding advances to unnamed parties

The liability and equity side, line by line

Line	What it is	What to check	Red flag
Equity share capital	Number of shares × face value. BWL: 12 crore × ₹10 = ₹120 crore. Changes only on issue, buyback, bonus or a split.	Track the share count over ten years, not the rupee figure	Steady dilution: a company whose share count rises 6% a year is transferring a sixth of your ownership away each decade
Other equity	Retained earnings, securities premium, general reserve, capital redemption reserve, OCI reserves, share-based payment reserve	The statement of changes in equity shows every movement. Retained earnings growth should roughly equal cumulative PAT less dividends less buybacks.	Unexplained direct charges to reserves that bypass the P&L
Borrowings (non-current and current)	Term loans, debentures, working capital facilities, commercial paper, external commercial borrowings, promoter loans	The note gives interest rates, security, repayment schedule and covenants. Build a maturity ladder. Note the split between fixed and floating rates, and any foreign-currency exposure.	**A large proportion of debt maturing within twelve months in a company with weak cash generation.** Refinancing risk is how solvent-looking companies fail. Also: short-term borrowings funding long-term assets.
Lease liabilities	The present value of future lease payments under Ind AS 116	**Include in debt.** A retailer with no borrowings but ₹3,000 crore of lease liabilities is a leveraged company.	Rating agencies have always treated leases as debt; equity investors frequently do not
Deferred tax liabilities / assets	Timing differences between book and tax treatment, chiefly depreciation	A large DTL is usually normal for a capital-intensive company. A large deferred tax *asset* is a bet on future taxable profits.	A DTA recognised on accumulated losses by a loss-making company is a profit booked on the assumption that losses will end

Trade payables	Money owed to suppliers	Payable days. The Schedule III ageing schedule, and the separate disclosure of amounts owed to micro and small enterprises under the MSMED Act, 2006.	**Stretching payables is a cheap but finite source of funding.** A sharp rise in payable days near year-end is window dressing. Overdue MSME dues attract mandatory interest and signal cash stress.
Provisions	Gratuity, leave encashment, warranty, litigation, decommissioning	The movement schedule: opening, additions, utilised, reversed, closing	Large reversals that boost profit; chronic under-provision for warranty in a product business
Contingent liabilities (*disclosed in the notes, not on the balance sheet*)	Disputed tax demands, guarantees given, claims not acknowledged as debts, pending litigation	**Always read this note.** Compare the total against equity. A company with ₹1,500 crore of equity and ₹1,100 crore of contingent liabilities has a materially different risk profile from the balance sheet alone.	**Corporate guarantees given on behalf of promoter group entities.** If the group entity defaults, the listed company pays. This has destroyed minority value repeatedly in India.

Working capital: where cash actually goes

Working capital is the money permanently locked in the operating cycle. It is a use of cash, it grows with revenue, and for a fast-growing business it can consume every rupee of profit.

TEXT

```
BWL WORKING CAPITAL – FY2025 (Rs crore)

OPERATING CURRENT ASSETS
  Inventories                              384
  Trade receivables                        432
  Other financial assets                    36
  Other current assets                      78
                                          ─────
                                           930

OPERATING CURRENT LIABILITIES
  Trade payables                           288
  Other financial liabilities               66
  Other current liabilities                 42
  Short-term provisions                     18
                                          ─────
                                           414

NET OPERATING WORKING CAPITAL            = 516
  (cash, bank and borrowings excluded – they are
   financing items, not operating ones)

As a % of revenue:  516 / 2,400 = 21.5%
FY2024 comparison:  486 / 2,120 = 22.9%  → improving

THE INCREMENTAL CALCULATION THAT MATTERS

  Revenue growth FY2025            = +280
  NWC growth FY2025                = + 30
  Incremental NWC / incremental
    revenue                        = 30 / 280 = 10.7%

  So every extra Rs 100 of revenue consumes about
  Rs 11 of additional working capital. With an EBIT
  margin of 15.5% and tax of 25%, every Rs 100 of
  extra revenue produces about Rs 11.6 of NOPAT.

  CONCLUSION: growth is roughly self-funding on the
  working-capital line. Capex is the binding
  constraint, not working capital.

  CONTRAST – a company where incremental NWC is 35%
  of incremental revenue and NOPAT margin is 8%:
  every Rs 100 of growth consumes Rs 35 and produces
  Rs 8. Growth is cash-negative. Such a company must
  borrow or issue equity to grow, and reported profit
  will never become cash. This is the single most
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common way a "high growth" Indian small-cap
destroys value.
```

> **Negative working capital is a competitive advantage**
> Some businesses collect from customers before paying suppliers: organised retail, quick-service restaurants, airlines selling tickets in advance, insurance, and subscription software billed annually. Their net operating working capital is **negative**, which means growth generates cash instead of consuming it — suppliers are funding the expansion for free. When you see a long run of negative working capital alongside growth, you are usually looking at genuine buyer power or a structurally favourable business model. It is also fragile: if growth stops, the float unwinds and cash flows out.

Reading the balance sheet as a system

Individual line items matter less than the relationships between them. Four checks, in order.

1. **Does the funding match the assets?**
   Long-lived assets should be funded by equity and long-term debt. A company funding a three-year plant construction with 90-day working capital lines is running a refinancing risk that has nothing to do with its profitability. Compare (equity + non-current liabilities) against non-current assets. BWL: 1,500 + 600 = 2,100 against non-current assets of 1,650 — comfortably funded, with ₹450 crore left over to cover net working capital.

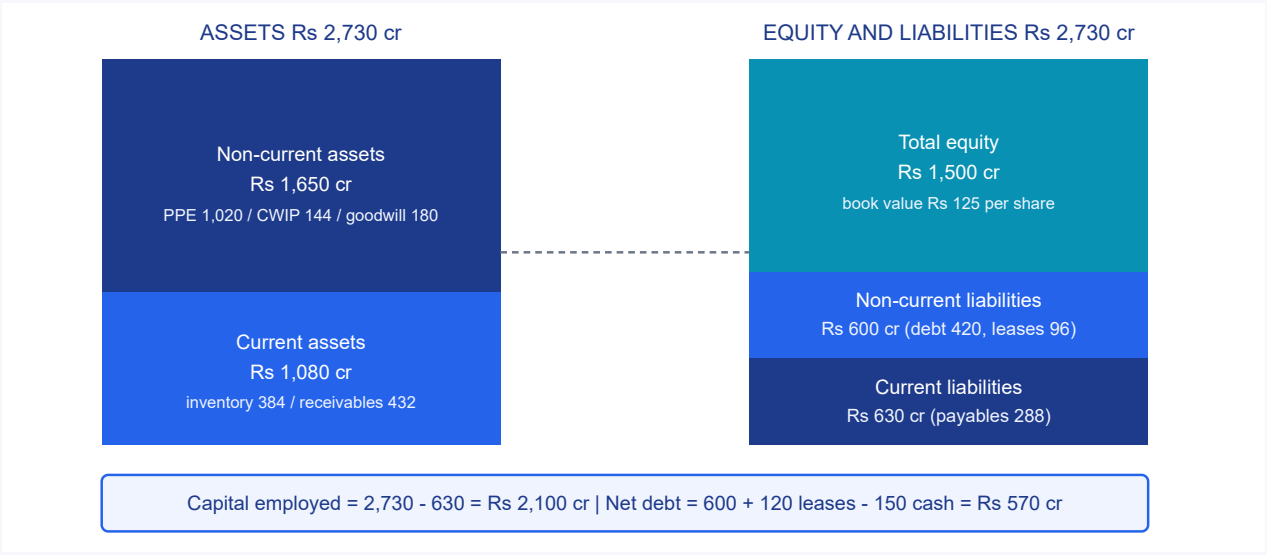
2. **Is growth in assets matched by growth in revenue?**
   Total assets grew 8.7% (2,512 to 2,730) while revenue grew 13.2%. Asset turnover is therefore improving. The opposite pattern — assets growing much faster than revenue for several years — means capital is being deployed into something that does not generate sales, and you must find out what.

3. **Does equity growth reconcile with retained profit?**
   Equity rose ₹180 crore (1,320 to 1,500). PAT was ₹240 crore, dividends ₹60 crore, OCI −₹2 crore. 240 − 60 − 2 = ₹178 crore, against ₹180 crore of actual increase; the small residual is the share-based payment reserve.
   It reconciles.
   When it does not reconcile, something has been charged directly to reserves, bypassing the P&L, and you need to find it in the statement of changes in equity.

4. **What happens in a bad year?**
   Apply a stress: revenue down 20%, EBITDA margin down 500 basis points. BWL's EBITDA would fall from ₹504 crore to roughly ₹1,920 × 16% = ₹307 crore. Against finance costs of ₹58 crore, interest coverage falls from 6.4× to about 3.0× on an EBIT basis — uncomfortable but

survivable. Repeat the same stress on a company with ₹2,000 crore of debt and the answer is a covenant breach.

Do this calculation before you buy, not after the bad year starts.



Bharat Widgets Ltd at 31 March 2025. Equity plus non-current liabilities of ₹2,100 crore comfortably funds non-current assets of ₹1,650 crore.

Schedule III disclosures most investors never read

⚠ **Book value is an accounting number, not an economic one**

BWL's book value per share of ₹125 is the historical cost of its assets less depreciation and liabilities. It systematically understates land bought decades ago and brands built over generations, and systematically overstates obsolete plant and goodwill from bad acquisitions. For a manufacturer it is a rough anchor. For an asset-light services or consumer company it is close to meaningless — a software firm's real assets walk out of the building every evening. Price-to-book is a useful ratio for banks and a misleading one almost everywhere else, for reasons set out in section 8.

The Cash Flow Statement

Profit is an opinion; cash is a fact. The cash flow statement reconciles the accounting profit in the P&L to the actual movement of money in the bank, and in doing so it exposes nearly every form of earnings management. If you only had time to read one

statement, read this one. If you only had time to compute one ratio, compute cumulative CFO divided by cumulative PAT over five to ten years.

The three sections and what each means

Section	What it captures	What a healthy mature business looks like	What a healthy young business looks like
Operating (CFO)	Cash generated by running the business: profit adjusted for non-cash items, then adjusted for the change in working capital, less tax actually paid	Consistently positive, consistently above PAT, growing with the business	Can legitimately be negative for a period while working capital builds — but the reason must be visible and bounded
Investing (CFI)	Capex, acquisitions, purchase and sale of investments, interest and dividends received	Negative, roughly equal to depreciation plus modest growth capex	Heavily negative as capacity is built
Financing (CFF)	Debt raised and repaid, equity issued, buybacks, dividends paid, interest paid, lease principal	Negative: the business returns cash to lenders and shareholders	Positive: the business is raising capital

✦ **The signature test**

Look at the signs of the three sections together over five years. **CFO positive, CFI negative, CFF negative** is a self-funding mature business returning cash — the most desirable pattern. **CFO positive, CFI very negative, CFF positive** is a business investing ahead of growth, which is fine if returns on that investment are above the cost of capital and visible within a few years. **CFO negative, CFI negative, CFF positive** for several consecutive years means the business is funded entirely by external capital and is burning it. In India that pattern has preceded a very large share of permanent capital losses.

BWL's cash flow statement, worked

Indian companies use the **indirect method**: start from PBT, strip out everything that is not cash, adjust for working capital, subtract tax paid.

TEXT

BHARAT WIDGETS LTD – CONSOLIDATED STATEMENT OF CASH FLOWS, YEAR ENDED 31 MARCH 2025 (Rs crore)

A. CASH FLOW FROM OPERATING ACTIVITIES	
Profit before tax	326
Adjustments for non-cash and non-operating:	
Depreciation and amortisation	132
Finance costs	58
Exceptional item (impairment, non-cash)	24
Loss on sale of property, plant and equip	4
Interest and dividend income	(36)
Operating profit before working capital changes	508
Changes in working capital:	
(Increase) in inventories	(24)
(Increase) in trade receivables	(36)
Increase in trade payables	36
(Increase) in other current assets	(4)
	(28)
Cash generated from operations	480
Direct taxes paid (net of refunds)	(98)
NET CASH FROM OPERATING ACTIVITIES (CFO)	382
B. CASH FLOW FROM INVESTING ACTIVITIES	
Purchase of PPE and additions to CWIP	(264)
Proceeds from sale of PPE	12
Purchase of non-current investments	(24)
Interest and dividends received	30
NET CASH USED IN INVESTING (CFI)	(246)
C. CASH FLOW FROM FINANCING ACTIVITIES	
Proceeds from long-term borrowings	120
Repayment of long-term borrowings	(126)
Payment of lease liabilities (principal and interest)	(34)
Finance costs paid	(56)
Dividend paid	(60)
NET CASH USED IN FINANCING (CFF)	(156)
NET CHANGE IN CASH (A + B + C)	(20)
Cash and cash equivalents, opening	116

```
Cash and cash equivalents, closing            96

TIE-OUT: closing cash of 96 matches the
balance sheet line exactly. It always must.
```

Why CFO is harder to fake than PAT

This is the central point of the entire section, and it is worth being precise about the mechanism rather than merely asserting it.

Manipulation	Effect on PAT	Effect on CFO	Why CFO is unaffected
Recognise revenue early / channel stuffing	Up	**Zero**	PBT rises, but receivables rise by the same amount, and the working-capital adjustment subtracts it straight back out
Under-provide for doubtful debts	Up	**Zero**	The provision is a non-cash charge; whether you book it or not, no cash moved
Extend asset useful lives	Up	**Zero**	Depreciation is added straight back as a non-cash item
Produce into inventory to absorb fixed overhead	Up	**Zero**	The inventory increase is subtracted in the working-capital adjustment
Reverse an old provision into profit	Up	**Zero**	Non-cash; added back
Capitalise operating expenses into CWIP	Up	**Up** — and this is the loophole	The spend moves out of CFO and into CFI as "capex". **This is why CFO alone is not enough and you must also look at free cash flow.**
Stretch payables at year-end	Zero	**Up**	A genuine, if temporary, cash benefit. Check payable days and whether it reverses in Q1.
Sell receivables / factoring and bill discounting	Small negative (the discount)	**Up**	Converts receivables to cash today. Legitimate, but it borrows from next year's CFO. Look for the disclosure of discounted bills.

✕ **The two loopholes in CFO, and how to close them**

CFO can be inflated by (a) **capitalising costs**, which pushes the outflow from CFO into CFI, and (b) **squeezing working capital at year-end** through payable stretching, aggressive collection or bill discounting. Both are closed by using **free cash flow** — CFO minus capex — and by checking that capex is proportionate to the capacity actually added. A company whose CFO is excellent but whose capex is three times depreciation with no new capacity to show for it has simply moved the problem one line down the statement.

Free cash flow: three definitions, and when to use which

TEXT

```
FREE CASH FLOW DEFINITIONS - BWL FY2025 (Rs crore)

1. SIMPLE FCF  (the most common, and fine for screening)
     CFO                                   382
     Less: total capex                    (264)
                                          ─────
     FCF                                   118

2. FCF TO EQUITY (FCFE) - what is actually available
   to shareholders after lenders are served
     CFO                                   382
     Less: capex                          (264)
     Add:  net debt raised  (120 - 126)     (6)
     Less: lease principal repaid          (22)
                                          ─────
     FCFE                                   90
     Dividend paid 60 → payout of FCFE = 67%.
     Sustainable.

3. FCF TO THE FIRM (FCFF) - what the whole business
   generates before any financing decision. This is
   the number the DCF in section 9 uses.
     EBIT                                  372
     Less: tax on EBIT at 25.17%           (94)
                                          ─────
     NOPAT                                 278
     Add:  depreciation and amortisation   132
     Less: capex                          (264)
     Less: increase in net working capital (30)
                                          ─────
     FCFF                                  116

MAINTENANCE vs GROWTH CAPEX - the judgement that
matters most and is never disclosed

     Total capex                           264
     Depreciation (a rough proxy for
       maintenance capex)                  132
                                          ─────
     Implied growth capex                  132

     "Owner earnings" on a maintenance-only basis:
       CFO 382 - maintenance capex 132 = Rs 250 cr
       = Rs 20.8 per share on a price of Rs 480
       = a 4.3% maintenance free cash flow yield

INTERPRETATION: BWL spends half its capex on growth.
If management stopped growing tomorrow, free cash
```

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flow would roughly double. Whether that would be a
good idea depends entirely on the return earned on
the growth capex — which is what incremental ROCE
in section 7 measures.
```

> **Depreciation is only a crude proxy for maintenance capex**
> In a high-inflation environment, replacing a machine bought ten years ago costs more than its historical-cost depreciation, so depreciation understates true maintenance capex. In a business with a shrinking asset base or rapidly cheapening technology, it overstates it. Better triangulation: look at capex in the years when the company explicitly says it is *not* expanding capacity, and read the capex guidance in the concall where management often splits maintenance from growth voluntarily.

The quality ratios, and what good looks like

Ratio	Formula	BWL FY2025	Strong	Investigate	W
CFO / EBITDA	Operating cash flow ÷ EBITDA	382 ÷ 504 = **75.8%**	Above 70%	Below 50% for two or more years	W ab op re in up
CFO / PAT	Operating cash flow ÷ profit after tax	382 ÷ 240 = **1.59×**	Above 1.0×, since depreciation is added back	Persistently below 1.0×	A pr cc ca
Cumulative CFO / cumulative PAT (10 yr)	Sum of ten years of each	**1.41×**	Above 1.0×	Below 0.8×	**Tl pe fc** O nc ye cc
FCF / revenue	(CFO − capex) ÷ revenue	118 ÷ 2,400 = **4.9%**	Above 5% for a manufacturer; above 15% for an asset-light business	Persistently negative	W bu ac pr sc
Capex / depreciation	Capex ÷ D&A	264 ÷ 132 = **2.0×**	1.0–1.5× for a mature business; above 2× for an expanding one	Above 2.5× for several years with no capacity or revenue growth	Ca ex pr de ex
Cash conversion	FCF ÷ PAT	118 ÷ 240 = **49%**	Above 60% for a mature business	Below 30% for several years	G cc er

Dividend / FCFE	Dividend paid ÷ FCFE	60 ÷ 90 = **67%**	Below 80%	Above 100%	A fu bo th op

Worked example: the company that looks profitable and burns cash

Here is a second invented company, **Sahyadri Infra Projects Ltd**, an EPC contractor. Its P&L is excellent. Its cash flow is a disaster. Every number below is internally consistent, and this precise pattern — strong reported profit, weak cash, ballooning receivables and unbilled revenue, funded by debt — has preceded a very large proportion of Indian corporate failures.

TEXT

SAHYADRI INFRA PROJECTS LTD (hypothetical)

Rs crore	FY21	FY22	FY23	FY24	FY25
PROFIT AND LOSS					
Revenue	840	1,060	1,360	1,720	2,180
Revenue growth	–	26%	28%	26%	27%
EBITDA	143	182	238	309	403
EBITDA margin	17.0%	17.2%	17.5%	18.0%	18.5%
Depreciation	34	42	52	64	78
Finance cost	46	62	84	116	158
PBT	63	78	102	129	167
PAT	47	58	76	96	124
PAT margin	5.6%	5.5%	5.6%	5.6%	5.7%
CASH FLOW					
CFO	22	11	-8	-34	-62
Capex	(68)	(86)	(112)	(148)	(196)
FCF	(46)	(75)	(120)	(182)	(258)
Net debt raised	58	94	142	206	288
BALANCE SHEET					
Trade receivables	294	420	598	842	1,178
Unbilled revenue (contract assets)	118	176	262	382	546
Inventories	84	112	152	208	282
Trade payables	148	196	264	352	468
Gross debt	398	492	634	840	1,128
Total equity	412	470	546	642	766
DERIVED RATIOS					
Receivable days	128	145	160	179	197
Receivables + unbilled as % of revenue	49%	56%	63%	71%	79%
CFO / EBITDA	15%	6%	-3%	-11%	-15%
Cumulative CFO (FY21–25)				-71	
Cumulative PAT (FY21–25)				401	
Cumulative CFO / PAT				-0.18x	
Net debt / EBITDA	2.6	2.6	2.6	2.6	2.7
Interest coverage (EBIT / finance cost)	2.4	2.3	2.2	2.1	2.1

Looks like: 27% revenue CAGR, expanding margin, consistently profitable. A screener would rank this stock highly on growth and consistency.

WHAT IS ACTUALLY HAPPENING

Revenue is being recognised on percentage-of-completion

```
as work is done, but customers are not paying. Unbilled
revenue – work done but not yet invoiced – has grown
from 14% of revenue to 25%. Receivables have grown from
35% to 54% of revenue. Together they now represent 79%
of a year's sales, up from 49%.

Over five years the company reported Rs 401 crore of
profit and consumed Rs 71 crore of operating cash. The
gap, plus Rs 610 crore of capex, has been funded
entirely by Rs 788 crore of fresh borrowing. Debt has
grown 183% while equity grew 86%.

Net debt / EBITDA looks stable at about 2.6x – which is
exactly the trap. It is stable only because EBITDA is
growing as fast as the debt, and EBITDA is growing
because of revenue that has not been collected.

THE SINGLE QUESTION THAT BREAKS THE THESIS

What happens when revenue growth stops? EBITDA stops
growing. Debt does not stop growing, because the
existing receivables still have to be funded. Net debt
/ EBITDA goes from 2.6x to 4x to 6x in three years
without the company doing anything differently. Then
the lenders reprice, the working capital lines are cut,
projects stall, receivables become disputed, and the
provision that has been deferred for five years arrives
all at once.

Nothing in the P&L predicts this. Everything in the
cash flow statement does.
```

Apply this test to every company you look at

Add up CFO for the last ten reported years. Add up PAT for the same ten years. If cumulative CFO is less than cumulative PAT, you need a specific, verifiable explanation — a genuine and bounded working-capital build during hypergrowth, a structural shift in the business model, a one-off. "It is a growing company" is not an explanation, because growth that permanently consumes more cash than it produces is not growth; it is a slow liquidation funded by lenders. This one calculation takes four minutes per company and removes a large share of the worst outcomes from your opportunity set.

Reconciling CFO to PAT in your head

PAT 240 → + tax 86 → PBT 326 → + non-cash 182 → − other income 36

→ + interest 58 → = 508 → − WC 28 → − tax paid 98 → CFO 382

Note the two places where real money is involved: the working-capital movement and the tax actually paid. Everything else is a reclassification. **Tax paid of ₹98 crore against a current tax charge of ₹92 crore** is a small, unremarkable difference arising from advance tax timing. A persistent and widening gap in the other direction — large tax charges in the P&L and small cash tax paid — means book profit and taxable profit have diverged, which is worth understanding through the tax reconciliation note.

Why the cash flow statement is only audited annually

Ratio Analysis in Full

A ratio is a compression device: it turns two numbers into one, so that companies of different sizes can be compared and so that a trend becomes visible. Every ratio also destroys information, and most have at least one sector where they are actively misleading. Learn the formula, the interpretation, the sector caveat, and what a deteriorating trend means — in that order.

⚠ **Four rules before any ratio**

One. A ratio in isolation means nothing; a ten-year trend and a peer comparison mean something. **Two.** Where a ratio divides a flow (from the P&L) by a stock (from the balance sheet), use the *average* of opening and closing balance sheet figures, not the closing figure — otherwise a company that raised equity in March will show an artificially depressed ROE. **Three.** Compute it yourself from the annual report; screener definitions vary and are frequently wrong. **Four.** Banks, NBFCs, insurers and holding companies need entirely different ratios — see section 12.

Profitability ratios

Ratio	Formula	BWL FY2025	What it measures	Sector caveat
Gross margin	(Revenue − direct material cost) ÷ revenue	1,272 ÷ 2,400 = **53.0%**	Pricing power and product mix, before any overhead	Not comparable across sectors at all. A 15% gross margin steel producer may be more profitable than a 60% gross margin apparel retailer once overheads are counted.
EBITDA margin	EBITDA ÷ revenue	504 ÷ 2,400 = **21.0%**	Operating profitability before capital structure, tax and the asset base	Misleading for capital-intensive businesses, where depreciation is a genuine cost. Inflated for lease-heavy businesses post Ind AS 116.
EBIT / operating margin	EBIT ÷ revenue	372 ÷ 2,400 = **15.5%**	True operating profitability including the cost of the asset base	Depends on depreciation policy, which varies between companies
Net profit margin	PAT ÷ revenue	240 ÷ 2,400 = **10.0%**	The bottom line as a share of sales	Contaminated by other income, one-offs, tax regime and leverage. Least useful of the margins for comparison.
ROE	PAT ÷ average shareholders' equity	240 ÷ ((1,500+1,320)÷2)	Return earned on the money shareholders	**Inflated by leverage.** A company can

		= 240 ÷ 1,410 = **17.0%**	have left in the business	raise ROE purely by borrowing. Always read ROE alongside debt-to-equity. Meaningless if equity is negative or tiny.
ROCE	EBIT ÷ average capital employed, where capital employed = total assets − current liabilities	372 ÷ ((2,100+1,944)÷2) = 372 ÷ 2,022 = **18.4%**	Return on all long-term capital, debt and equity together, before financing effects	The best single profitability ratio for a non-financial company, because it is capital-structure neutral. Definitions of "capital employed" vary; state yours.
ROIC	NOPAT ÷ invested capital, where NOPAT = EBIT × (1 − tax rate) and invested capital excludes surplus cash	278 ÷ 2,154 = **12.9%** (see note below)	After-tax return on capital genuinely at work in the business	The most economically meaningful and the least standardised. The definition of invested capital drives the answer.

TEXT

```
WHY ROCE AND ROIC DISAGREE - BWL FY2025

ROCE  = EBIT / average capital employed
      = 372 / 2,022  =  18.4%   (pre-tax)

ROIC  = NOPAT / invested capital
NOPAT = 372 x (1 - 0.2517)  =  278

Invested capital, definition A (top-down):
  Total assets                          2,730
  Less: cash and bank                    (150)
  Less: non-interest-bearing current
        liabilities (288+66+42+18+12)    (426)
                                        -----
                                        2,154
  ROIC = 278 / 2,154  =  12.9%

Invested capital, definition B (bottom-up):
  Net PPE                               1,020
  CWIP                                    144
  Right-of-use assets                     108
  Goodwill and intangibles                276
  Net operating working capital           516
  Other non-current assets                 42
  Less: long-term provisions              (30)
  Less: deferred tax liability            (54)
                                        -----
                                        2,022
  ROIC = 278 / 2,022  =  13.7%

THREE LESSONS
1. ROCE is PRE-TAX, ROIC is POST-TAX. Comparing them
   directly is an error. BWL's pre-tax ROCE of 18.4%
   becomes 13.8% post-tax, which is the number to
   compare against a ~12.5% WACC.
2. Including or excluding CWIP changes the answer a
   lot. CWIP is capital invested but not yet earning.
   Excluding it (2,022 - 144 = 1,878) gives ROIC of
   14.8% - the return on capital actually WORKING.
   Both numbers are useful: one asks "how good is the
   business", the other asks "how good is the business
   including the capital you have tied up waiting".
3. Goodwill of 180 is capital that was spent. Excluding
   it, as some analysts do, would give ROIC of
   278/1,842 = 15.1%. That measures operating quality
   but ignores the price paid for acquisitions. If
   management keeps acquiring, you must include it.
```

```
RULE: state your definition, apply it identically to
every peer, and never mix your number with someone
else's.
```

> **The number that actually decides whether growth creates value**
>
> Compare post-tax ROIC against WACC. BWL earns roughly 13.8% post-tax on invested capital against a cost of capital of about 12.5% (derived in section 9). The spread is **+1.3 percentage points**. That is positive but thin — which means growth creates value, but only just, and a modest deterioration in either number would make growth value-destructive. A company earning 25% ROIC against a 12% WACC should reinvest every rupee it can. A company earning 8% against a 12% WACC should stop investing and return the cash, and a management team that keeps expanding at 8% is destroying your capital at a rate of 4% of every rupee invested, no matter how impressive the revenue growth chart looks.

DuPont decomposition, worked three ways

DuPont analysis breaks ROE into its drivers so that you can see *why* it is what it is, and which component is changing. There are three standard versions of increasing granularity.

TEXT

```
DUPONT ANALYSIS - BHARAT WIDGETS LTD FY2025
(all balance sheet items at closing values for clarity;
 use averages in production work)

----- VERSION 1: THREE-STEP -----

  ROE = Net margin x Asset turnover x Equity multiplier

  Net margin       = PAT / Revenue
                   = 240 / 2,400          = 0.1000
  Asset turnover   = Revenue / Total assets
                   = 2,400 / 2,730        = 0.8791
  Equity multiplier= Total assets / Equity
                   = 2,730 / 1,500        = 1.8200

  ROE = 0.1000 x 0.8791 x 1.8200          = 0.1600
                                          = 16.0%

  Cross-check: 240 / 1,500 = 16.0%.  OK.

  READING: BWL earns a 10% margin, turns its asset
  base over 0.88 times a year, and uses modest
  leverage of 1.82x. None of the three is extreme.
  This is a balanced profile.

----- VERSION 2: FIVE-STEP -----

  ROE = Tax burden x Interest burden x Operating
        margin x Asset turnover x Equity multiplier

  Tax burden       = PAT / PBT
                   = 240 / 326            = 0.7362
  Interest burden  = PBT / EBIT*
                   = 326 / 384            = 0.8490
  Operating margin = EBIT* / Revenue
                   = 384 / 2,400          = 0.1600
  Asset turnover   = 2,400 / 2,730        = 0.8791
  Equity multiplier= 2,730 / 1,500        = 1.8200

  * EBIT here = PBT + finance costs = 326 + 58 = 384,
    which includes other income and is AFTER the
    exceptional item. This is the standard DuPont
    convention and differs from the 372 used for
    ROCE. Pick one convention and hold it.

  ROE = 0.7362 x 0.8490 x 0.1600 x 0.8791 x 1.8200
      = 0.1600 = 16.0%.  OK.
```

```
  READING, component by component:
    Tax burden 0.736  → an effective tax rate of
      26.4%, close to the statutory 25.17%. No tax
      holiday propping up the result. Durable.
    Interest burden 0.849 → interest and the
      exceptional item consume 15% of operating
      profit. Moderate. A number below 0.70 means
      debt is eating the business.
    Operating margin 0.160 → the operating quality
      of the business. This is where a moat shows up.
    Asset turnover 0.879 → capital intensity.
    Equity multiplier 1.82 → leverage.

──── VERSION 3: FIVE-STEP TREND, FY2021 to FY2025 ────
  (this is the version that is actually useful)

                     FY21   FY22   FY23   FY24   FY25
  Tax burden        0.734  0.738  0.741  0.740  0.736
  Interest burden   0.772  0.794  0.818  0.836  0.849
  Operating margin  0.139  0.147  0.152  0.159  0.160
  Asset turnover    0.798  0.836  0.845  0.844  0.879
  Equity multiplier 2.105  2.034  1.961  1.903  1.820
                    ─────  ─────  ─────  ─────  ─────
  ROE               13.2%  15.2%  15.9%  16.9%  17.0%
  (computed on average equity; the product of the
   five closing-value factors gives 16.0%, and the
   difference is purely the averaging convention)

  DECOMPOSING THE +3.8 POINT IMPROVEMENT:

    Operating margin  0.139 → 0.160  CONTRIBUTED +
      The business genuinely became more profitable.
    Interest burden   0.772 → 0.849  CONTRIBUTED +
      Debt was repaid; less of operating profit is
      going to lenders.
    Asset turnover    0.798 → 0.879  CONTRIBUTED +
      The asset base is being used harder.
    Equity multiplier 2.105 → 1.820  CONTRIBUTED -
      Leverage FELL, which reduces ROE.
    Tax burden        flat             NEUTRAL

  THE CONCLUSION THAT MATTERS: ROE rose 3.8 points
  DESPITE deleveraging. Three operating factors all
  improved while the one financial-engineering factor
  worked against the company. This is the highest
  quality form of ROE improvement there is.

  CONTRAST - the low-quality version of the same
  headline. Suppose instead:
```

```
  Operating margin  0.139 → 0.128  (deteriorating)
  Asset turnover    0.798 → 0.762  (deteriorating)
  Equity multiplier 2.105 → 3.400  (rising debt)
ROE would still rise to about 17%, and a screener
sorting on ROE would rank the two companies
identically. One is improving; the other is one
bad year from a covenant breach. DuPont is how you
tell them apart in sixty seconds.
```

✦ **Use DuPont on every ROE change you cannot explain**

Whenever a company's ROE moves by more than two points in either direction, decompose it before forming a view. Roughly four outcomes: margin-driven (investigate whether it is pricing, mix or cost), turnover-driven (usually asset sweating or a working-capital change, and check whether capacity is now constrained), leverage-driven (the lowest-quality source, and reversible in the worst way), or tax-driven (check whether a holiday is expiring). The decomposition takes two minutes and changes the interpretation completely.

Efficiency ratios: the cash conversion cycle

TEXT

```
WORKING CAPITAL EFFICIENCY - BWL FY2025

Use AVERAGE balances. Opening and closing:
  Inventories        360 --> 384   average  372
  Trade receivables  396 --> 432   average  414
  Trade payables     252 --> 288   average  270

Cost of goods sold (for inventory and payable days):
  Cost of materials consumed       1,104
  + Purchases of stock-in-trade       72
  + Changes in inventories FG/WIP    (48)
                                   ------
  COGS                              1,128

INVENTORY DAYS
  = (average inventory / COGS) x 365
  = (372 / 1,128) x 365   =  120.4 days

RECEIVABLE DAYS (DSO)
  = (average receivables / revenue) x 365
  = (414 / 2,400) x 365   =   63.0 days

PAYABLE DAYS (DPO)
  = (average payables / COGS) x 365
  = (270 / 1,128) x 365   =   87.4 days

CASH CONVERSION CYCLE
  = inventory days + receivable days - payable days
  = 120.4 + 63.0 - 87.4
  =  96.0 days

WHAT 96 DAYS MEANS IN RUPEES
  Daily revenue        = 2,400 / 365  = Rs 6.58 cr
  Cash locked up       = 96 x 6.58    = Rs 632 cr
  (close to the Rs 516 cr net operating working
   capital computed earlier; the gap is because the
   CCC uses COGS for two legs and revenue for one)

  At a 9% cost of debt, 96 days of working capital
  costs roughly 516 x 0.09 = Rs 46 crore a year -
  which is 12% of PAT.

THE SENSITIVITY THAT MANAGEMENT CARES ABOUT
  Reducing inventory days from 120 to 100 releases
    (20/365) x 1,128 = Rs 62 crore of cash, ONCE.
  Reducing receivable days from 63 to 55 releases
    (8/365) x 2,400  = Rs 53 crore, ONCE.
  Together: Rs 115 crore, or 44% of a year's capex,
```

```
funded from the balance sheet rather than the bank.

This is why "working capital release" appears in
every turnaround story, and why it is a ONE-TIME
benefit that cannot be repeated.
```

Business type	Typical inventory days	Typical receivable days	Typical payable days	Typical CCC	W
FMCG with strong brands	30–55	10–25	45–90	−20 to +20	D or ac cc di te
Organised retail / QSR	25–60	2–6	50–90	−40 to −10	Ca th or N fu rc
IT services	0 (no inventory)	55–80	20–35	30–55	Bi w er cl ur
Pharmaceutical formulations	90–160	70–120	60–100	80–170	Lo m ar cy di cr re
Auto components (OEM supply)	35–70	55–85	60–95	40–70	O ov bu di te
Capital goods / EPC	80–180	110–220	90–160	110–240	Lo cy m ba di
Jewellery retail	130–220	3–15	25–70	90–180	Er in

					bu sa ca
Commodity chemicals	45–80	50–80	45–75	50–90	St in

⚠ **Never compare a cash conversion cycle across sectors**

A jewellery retailer with a 170-day CCC is not worse-managed than a QSR chain with a −30-day CCC; they are different businesses. The comparison that carries information is **the same company against its own history** and **against its direct competitors**. A ten-year chart of CCC for a company and its three closest peers is one of the most revealing pictures in all of fundamental analysis, because deterioration shows up there months before it shows up in margins.

Leverage and solvency ratios

Ratio	Formula	BWL FY2025	Comfortable	Stressed	N
Debt / equity	Total borrowings ÷ total equity	600 ÷ 1,500 = **0.40** Incl. leases: 720 ÷ 1,500 = **0.48**	Below 0.5 for a manufacturer; below 1.0 for a stable utility	Above 1.5 for a cyclical business	D to ar co M ba N le 1(bu
Net debt / EBITDA	(Debt − cash) ÷ EBITDA	450 ÷ 504 = **0.89×** Incl. leases: 570 ÷ 504 = **1.13×**	Below 2.0×	Above 3.5× for a cyclical, above 4.5× for a stable business	**Th in le** It m op w re Le co us th
Interest coverage	EBIT ÷ finance costs	372 ÷ 58 = **6.4×**	Above 4×	Below 2.5×	Th te th se ou op Be co bo pa
Debt service coverage	EBITDA ÷ (interest + principal repayments due)	504 ÷ (58 + 126) = **2.74×**	Above 1.5×	Below 1.2×	In pr in co ig

					co ha in co st
Debt / capital employed	Total debt ÷ (debt + equity)	600 ÷ 2,100 = **28.6%**	Below 40%	Above 60%	U co ca st co di
Short-term debt / total debt	Current borrowings ÷ total borrowings	180 ÷ 600 = **30%**	Below 40%	Above 60%	Re Th m tw w co cr re to

Liquidity ratios

Ratio	Formula	BWL FY2025	Interpretation	Caveat
Current ratio	Current assets ÷ current liabilities	1,080 ÷ 630 = **1.71**	Above 1.0 means short-term assets cover short-term obligations on paper	A high current ratio can mean bloated inventory and uncollectable receivables. A ratio of 3.0 is often *worse* than 1.5. Read the composition.
Quick ratio (acid test)	(Current assets − inventories) ÷ current liabilities	(1,080 − 384) ÷ 630 = 696 ÷ 630 = **1.10**	Removes the least liquid current asset	Still includes receivables, which may be no more collectible than inventory is saleable
Cash ratio	(Cash + bank) ÷ current liabilities	150 ÷ 630 = **0.24**	The most conservative test: what could be paid today	Below 0.2 is normal for a company with committed bank lines. Check the undrawn facilities disclosure.

Liquidity ratios are the weakest family, and here is why

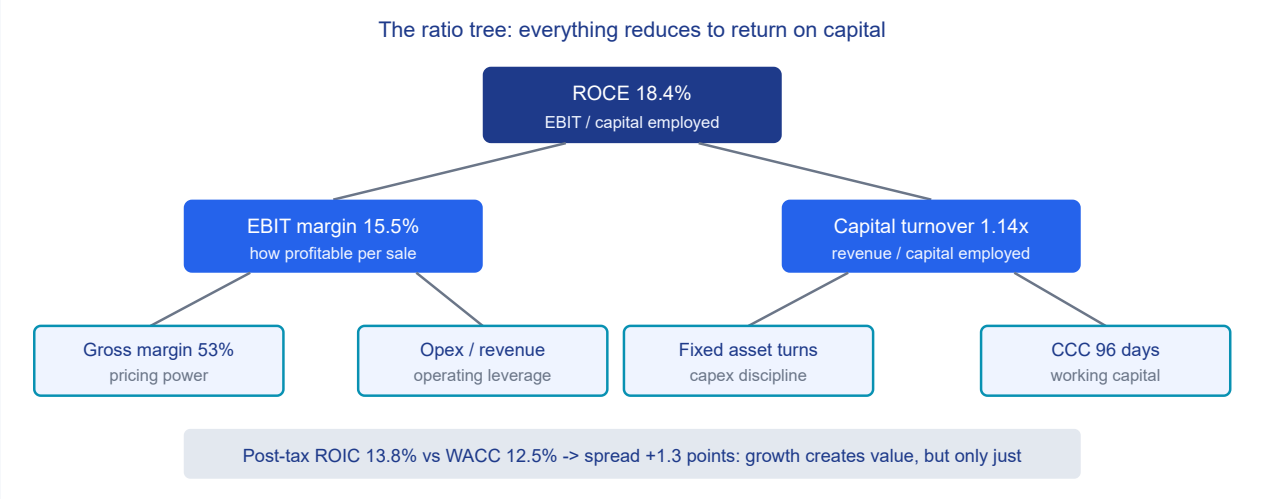
They are point-in-time measures of a continuous process, taken on the one day of the year the company knows you will look at. They can be managed upwards in the final fortnight by delaying supplier payments, pushing collections and drawing down a term loan. They also treat all current assets as equally liquid, which is plainly false. For a genuine view of short-term solvency, use the **cash flow statement** plus the **debt maturity schedule** in the borrowings note plus the **undrawn credit facilities** disclosure. Those three together tell you what the current ratio pretends to.

The complete ratio dashboard for BWL

Category	Ratio	FY2021	FY2022	FY2023	FY
Profitability	Gross margin	52.0%	51.6%	52.1%	52
	EBITDA margin	18.9%	19.4%	19.8%	20
	ROE (avg equity)	13.2%	15.2%	15.9%	16
	ROCE (avg)	14.6%	16.1%	16.9%	17
Efficiency	Inventory days	138	131	128	12
	Receivable days	71	69	67	65
	Cash conversion cycle	122	114	107	10
Leverage	Net debt / EBITDA	2.04×	1.72×	1.38×	1.
	Interest coverage	3.3×	4.0×	4.8×	5.
	Debt / equity	0.74	0.65	0.56	0.
Cash quality	CFO / EBITDA	66%	71%	73%	74
	CFO / PAT	1.43×	1.48×	1.52×	1.

	Capex / depreciation	1.4×	1.6×	1.7×	1.

A dashboard like this takes perhaps ninety minutes to build from ten annual reports and it is the backbone of every judgement that follows. Notice that every family of ratios is moving in the same direction and that the improvement is operating rather than financial. **Consistency across unrelated ratio families is itself evidence** — manipulation is usually localised, so a company where margins improve while cash conversion deteriorates and receivable days lengthen is telling a story that does not hold together.



Every profitability ratio decomposes into margin and capital turnover. Both decompose further into operating drivers you can actually investigate.

Growth Analysis

Growth is not automatically good. Growth funded at a return below the cost of capital destroys value faster the quicker it happens. The questions that matter are: where did the growth come from, what did it cost in capital, is it repeatable, and what return is the incremental capital earning? A revenue CAGR chart answers none of these.

Measuring growth properly

TEXT

```
BWL GROWTH ARITHMETIC

COMPOUND ANNUAL GROWTH RATE
  CAGR = (ending / beginning)^(1/years) - 1

  Revenue FY2016 = 1,012    FY2025 = 2,400
  9-year CAGR = (2,400/1,012)^(1/9) - 1
              = (2.3715)^(0.11111) - 1
              = 1.1010 - 1   =  10.1%

  PAT FY2016 = 63 (6.2% of 1,012)
  PAT FY2025 = 240
  9-year CAGR = (240/63)^(1/9) - 1
              = (3.8095)^(0.11111) - 1
              = 1.1600 - 1   =  16.0%

  EPS FY2016 = 63/11.4 cr shares = Rs 5.53
  EPS FY2025 = 240/12.0 cr shares = Rs 20.00
  9-year CAGR = (20.00/5.53)^(1/9) - 1  =  15.4%

THE GAP THAT MATTERS
  Revenue CAGR 10.1%  →  PAT CAGR 16.0%
  The 5.9-point gap is MARGIN EXPANSION, which is
  finite. Margins cannot expand forever. Over the
  next decade, PAT growth will converge towards
  revenue growth unless margins keep rising.

  EPS CAGR 15.4% < PAT CAGR 16.0% because the share
  count rose from 11.4 to 12.0 crore (ESOP exercise).
  ALWAYS CHECK THIS GAP. A company growing PAT at 20%
  while issuing 8% more shares a year is growing your
  per-share claim at about 11%.

THE BASE-EFFECT TRAP
  FY2020 revenue 1,368 (a weak year)
  FY2021 revenue 1,490  →  +8.9%
  FY2022 revenue 1,742  →  +16.9%

  A headline of "17% growth" in FY2022 is partly a
  recovery from a depressed base. The honest measure
  is the 2-year CAGR off FY2020, or better, the CAGR
  off the pre-disruption FY2019 peak of 1,402:
    (1,742/1,402)^(1/3) - 1  =  7.5% per year
  That is the real underlying rate, and it is less
  than half the headline.

  RULE: in any year following an abnormal base, quote
```

```
a multi-year CAGR alongside the year-on-year figure.
Companies quote whichever flatters them.
```

Organic versus inorganic growth

Source of growth	How to detect it	Quality	What to check
Volume growth	Company discloses tonnes, units or subscribers in the investor presentation	**Highest.** Real market demand being served	Compare against industry volume growth from an independent association. Growing faster means share gain.
Price / realisation growth	Revenue growth exceeds volume growth	**High if sustainable.** Evidence of pricing power	Is it genuine pricing power, or just passing on input-cost inflation? Check whether gross margin held.
Mix improvement	Gross margin rising with flat volume and price	**High**	Shift towards premium products or higher-margin geographies. Usually discussed on the call.
Acquisition	Goodwill appears or increases; the cash flow statement shows "acquisition of subsidiary, net of cash"; a note describes the business combination	**Variable, often poor.** Most acquisitions destroy value for the acquirer	**Strip it out.** Compute organic growth = reported growth minus the acquired entity's revenue. Then check the price paid against the acquired profit, and watch for impairment three years later.
New capacity	CWIP converting to PPE; capacity disclosures	**Good if returns hold**	Incremental ROCE on the new capacity. A new plant that earns 8% when the existing business earns 18% drags the whole company down.
Accounting change	A note on a change in accounting policy or a restatement of the prior year	**Not growth at all**	Gross-versus-net presentation changes, consolidation of a previously unconsolidated entity, Ind AS 115 transition effects

Currency	For exporters, revenue in rupees rises because the rupee depreciated	**Not operating growth**	Constant-currency growth is usually disclosed by IT and pharma companies. Use it.

TEXT

```
SEPARATING ORGANIC FROM INORGANIC - WORKED

Hypothetical: Konkan Precision Ltd, FY2025
  Reported revenue FY2025            1,840
  Reported revenue FY2024            1,420
  Reported growth                    29.6%

  Note 41 (business combinations) states the company
  acquired Malwa Components Pvt Ltd on 1 July 2024
  for Rs 310 crore, and Malwa contributed Rs 285
  crore of revenue for the 9 months consolidated.

  ORGANIC REVENUE FY2025 = 1,840 - 285 = 1,555
  ORGANIC GROWTH = 1,555 / 1,420 - 1 = 9.5%

  So "29.6% growth" is 9.5% organic plus 20.1%
  acquired. A completely different company from the
  one the headline describes.

  NOW CHECK WHETHER THE ACQUISITION WAS ANY GOOD
    Price paid                        310
    Malwa annualised revenue          380  (285 x 4/3)
    Malwa annualised EBITDA            49
    EV / EBITDA paid            310/49 = 6.3x
    Goodwill recognised               118
    Net identifiable assets acquired  192

    Implied pre-tax return on the purchase price:
      Malwa EBIT (say 49 - 14 depreciation)  = 35
      35 / 310  =  11.3% pre-tax
      Post-tax at 25.17%: 8.5%

    Against a WACC of roughly 12%, this acquisition
    DESTROYS value at the price paid, even though it
    adds Rs 285 crore of revenue and some profit.

  WHAT TO WATCH NEXT: whether the Rs 118 crore of
  goodwill survives the annual impairment test in
  FY2027 and FY2028. Goodwill impairments are the
  market's delayed acknowledgement that a price was
  too high, and they arrive two to four years later.
```

Operating leverage: the mathematics

TEXT

```
OPERATING LEVERAGE - WHY MARGINS MOVE MORE THAN
REVENUE

A company's costs split into variable (move with
volume) and fixed (do not). BWL's rough split:

  Variable costs (materials, freight,
    power, packing)                  1,310   54.6%
  Fixed costs (employee, rent,
    admin, selling, insurance)         586   24.4%
  ------------------------------------------
  Total operating cost (ex-D&A)      1,896   79.0%
  EBITDA                               504   21.0%

DEGREE OF OPERATING LEVERAGE (DOL)
  DOL = contribution / EBITDA
      = (revenue - variable costs) / EBITDA
      = (2,400 - 1,310) / 504
      = 1,090 / 504
      = 2.16x

  INTERPRETATION: a 1% change in revenue produces a
  2.16% change in EBITDA, in either direction.

TEST IT - REVENUE UP 10%
  Revenue               2,400 -> 2,640   (+10%)
  Variable costs        1,310 -> 1,441   (+10%)
  Fixed costs             586 ->   586   (flat)
  EBITDA                  504 ->   613   (+21.6%)
  EBITDA margin         21.0% -> 23.2%
  Check: 10% x 2.16 = 21.6%.  Confirmed.

TEST IT - REVENUE DOWN 10%
  Revenue               2,400 -> 2,160
  Variable costs        1,310 -> 1,179
  Fixed costs             586 ->   586
  EBITDA                  504 ->   395   (-21.6%)
  EBITDA margin         21.0% -> 18.3%

TEST IT - REVENUE DOWN 25% (a real recession)
  Revenue                      -> 1,800
  Variable costs               ->   983
  Fixed costs                  ->   586
  EBITDA                       ->   231   (-54%)
  EBITDA margin                -> 12.8%
  EBIT = 231 - 132 depreciation =    99
  Interest coverage = 99 / 58   =  1.7x
  -> Covenant territory. This is the stress number
```

```
      you should know BEFORE you buy.

COMBINING OPERATING AND FINANCIAL LEVERAGE
  Degree of financial leverage = EBIT / (EBIT - interest)
                               = 372 / (372 - 58) = 1.18x
  Total leverage = 2.16 x 1.18 = 2.55x
  A 10% revenue fall produces a 25.5% fall in PBT.

  For a company with DOL of 4 and DFL of 2.5, total
  leverage is 10x: a 10% revenue fall wipes out PBT
  entirely. Airlines, steel, shipping and hotels sit
  in this region. This is the arithmetic behind why
  cyclicals are dangerous, and it has nothing to do
  with the quality of management.
```

Operating leverage cuts both ways and is not a quality signal

High operating leverage makes good years spectacular and bad years catastrophic. It is a structural property of the cost base, not evidence of a good business. When you read that a company's margins "expanded strongly on operating leverage", the correct response is to compute what happens when revenue falls by the amount it fell in the last downturn. For BWL, FY2020 saw revenue fall 2.4% and EBITDA margin fall 180 basis points — consistent with a DOL of roughly 2 at that time. The arithmetic is honest even when the narrative is not.

Incremental ROCE: the most under-used growth metric

TEXT

```
INCREMENTAL RETURN ON CAPITAL EMPLOYED

The question: what return did the company earn on the
capital it added over the period? This is far more
informative than the average ROCE, because the average
is dominated by legacy assets that cannot be changed.

  Incremental ROCE
    = (EBIT_end - EBIT_start) / (CE_end - CE_start)

BWL, FY2021 to FY2025
  EBIT FY2021                 =   192
  EBIT FY2025                 =   372
  Change in EBIT              =   180

  Capital employed FY2021     = 1,586
  Capital employed FY2025     = 2,100
  Change in capital employed  =   514

  Incremental ROCE = 180 / 514  =  35.0%  (pre-tax)
  Post-tax at 25.17%            =  26.2%

  Average ROCE over the period  =  ~17%

  READING: the capital BWL deployed over the last
  four years earned roughly twice the return of its
  legacy capital base. This is why ROCE has been
  rising, and it is a strong argument for the company
  continuing to reinvest rather than pay out.

CONTRAST - a value-destroying grower
  EBIT change over 4 years           +  96
  Capital employed change            + 980
  Incremental ROCE = 96/980   =  9.8% pre-tax
                              =  7.3% post-tax
  Against a WACC of 12%, every rupee of growth capex
  destroys about 4.7 paise a year, in perpetuity. The
  revenue chart still goes up and to the right.

THE SUSTAINABLE GROWTH RATE
  g = ROE x retention ratio
    = ROE x (1 - dividend payout)

  BWL: ROE 17.0%, dividend 60 on PAT 240
       payout = 25%, retention = 75%
  g = 0.170 x 0.75  =  12.75%

  INTERPRETATION: BWL can grow earnings at about
```

12.75% a year indefinitely WITHOUT raising external capital or increasing leverage. Its actual PAT growth has been ~16%, which is only possible because margins were expanding. Once margins plateau, 12.75% is the ceiling.

This single calculation is the sanity check on every growth forecast you will ever read. If an analyst projects 25% earnings growth for a company with 17% ROE and a 25% payout, they are implicitly assuming either margin expansion, an equity raise, rising leverage, or an acquisition. Make them say which.

The growth-versus-quality trade-off

Profile	Revenue growth	ROCE	Typical valuation	What has to go right	W g
High growth, high ROCE	20%+	25%+	Very expensive (35–70× earnings)	The runway must be long enough to justify the multiple, and returns must not decay as scale increases	G de ye th ex de br bo m or
High growth, low ROCE	20%+	8–12%	Often expensive anyway, on price-to-sales	ROCE must improve with scale, which requires a specific, identifiable mechanism	It G co le ar is sc fa **m da ca In ca**
Low growth, high ROCE	5–10%	25%+	Moderate (15–25×)	Excess cash must be returned to shareholders rather than reinvested at low returns or spent on diversification	M "c ar bu re de va bu
Low growth, low ROCE	0–6%	Under 10%	Cheap (6–12×)	Either a cyclical recovery, an asset-value realisation, or a	It a cl tr

				genuine turnaround	ev ar
Cyclical at the peak	30%+ (temporarily)	30%+ (temporarily)	Looks cheapest exactly at the top, because E is at a cyclical maximum	Nothing — the cycle will turn	Bu ch co pe w 40 ea ye **cy P/ si hi of si**

> ✦ **The one growth question worth asking first**
> Before analysing any growth company, ask: **what is the mechanism by which this growth continues, and what would stop it?** Acceptable answers are specific: an under-penetrated category with a measurable penetration rate, a capacity expansion that is already funded and under construction, a regulatory change with a known implementation date, a distribution network being extended into named geographies. Unacceptable answers are "the industry is expected to grow at 18% CAGR" (that is a consultancy's forecast, not a mechanism) and "management is confident". If you cannot name the mechanism, you do not have a growth thesis; you have an extrapolation.

Valuation Part 1 — Relative

Relative valuation asks what similar businesses trade at and applies that multiple to the one in front of you. It is fast, intuitive, universally used, and quietly circular — it tells you what a company is worth *if the market is correctly valuing its peers*. It cannot tell you whether an entire sector is mispriced. Used with that limitation understood, it is the most practical valuation tool available.

The multiples, in order of usefulness

Multiple	Formula	BWL at ₹480	Best for	Breaks down when
P/E (trailing)	Price ÷ trailing twelve-month diluted EPS	480 ÷ 19.84 = **24.2×**	Stable, profitable, moderately leveraged companies	Earnings are negative, cyclical, dominated by one-offs, or the capital structures of the peers differ widely
P/E (forward)	Price ÷ estimated forward EPS	480 ÷ 23.10 = **20.8×** (on an estimate)	Everything — markets price the future, not the past	The estimate is someone else's guess. Always know whose, and what growth it assumes.
EV / EBITDA	(Market cap + net debt) ÷ EBITDA	(5,760 + 570) ÷ 504 = **12.6×**	Comparing companies with different leverage, tax positions or depreciation policies. The default for capital-intensive sectors.	Capital intensity differs sharply between the peers — EBITDA ignores the capex needed to sustain it. Post-Ind AS 116 lease treatment distorts cross-period comparison.
EV / EBIT	EV ÷ EBIT	6,330 ÷ 372 = **17.0×**	The same as EV/EBITDA but honest about depreciation. Underused.	Depreciation policies differ materially between peers
P/B	Price ÷ book value per share	480 ÷ 125 = **3.84×**	**Banks, NBFCs and insurers**, where the balance sheet *is* the business and assets are marked near fair value	Asset-light businesses, anything with large goodwill or brands, companies with old land at historical cost. For

				BWL it is near-meaningless.
EV / Sales	EV ÷ revenue	6,330 ÷ 2,400 = **2.64×**	Loss-making or early-stage companies where earnings are not yet meaningful	Always, really. It implicitly assumes a margin. Two companies on 2× sales with 5% and 25% margins are not comparably valued.
P/S	Market cap ÷ revenue	5,760 ÷ 2,400 = **2.40×**	Rarely better than EV/Sales, because it ignores debt	Any company with meaningful leverage
PEG	P/E ÷ expected earnings growth rate (in percentage points)	24.2 ÷ 14 = **1.73**	A rough screen for whether a high multiple is justified by growth	**Very easily gamed.** The growth rate is an input you choose. It ignores risk, capital intensity and growth duration. A company growing 40% for two years and a company growing 15% for fifteen years can have the same PEG and wildly different values.
Dividend yield	Dividend per share ÷ price	5.00 ÷ 480 = **1.04%**	Mature, cash-generative companies where payout is the primary return	Growth companies that correctly pay nothing. A very high yield is usually a market forecast that the dividend will be cut.
FCF yield	Free cash flow per share ÷ price	(118 ÷ 12) ÷ 480 = 9.83 ÷ 480 = **2.05%**	**Arguably the single best multiple**,	Capex is lumpy — use a three-to-five year average.

		On maintenance FCF: 20.83 ÷ 480 = **4.34%**	because it uses cash rather than accounting profit	Distinguish maintenance from growth capex.
Market cap / sales	Same as P/S	2.40×	Quick screening only	See EV/Sales

Building a peer comparison table properly

This is the core deliverable of relative valuation. Five hypothetical engineering-component companies, all invented, with BWL in the middle.

Company (all hypothetical)	Mkt cap (₹ cr)	Net debt (₹ cr)	EV (₹ cr)	Revenue	EI
Bharat Widgets	5,760	570	6,330	2,400	21
Deccan Components	12,400	−340	12,060	3,650	24
Konkan Precision	4,180	1,460	5,640	1,840	19
Vidarbha Engineering	2,920	410	3,330	1,520	17
Malwa Industries	1,640	880	2,520	1,960	11
Median	—	—	—	—	19

HOW TO ACTUALLY READ THAT TABLE

- **Malwa at 9.1× earnings is not cheap.** It has the lowest margin, the lowest ROCE, the slowest growth, the worst cash conversion and leverage of nearly 4× EBITDA. The multiple is low because the business is poor. **This is what "low P/E for a reason" looks like in a table.**
- **Deccan at 31.4× is not obviously expensive.** It has the highest margin, the highest ROCE, net cash, the best cash conversion and strong growth. Whether 31× is too much depends on how long that superiority persists — which is a judgement about the moat, not a judgement about the multiple.
- **Konkan looks cheap on P/E (19.8×) and expensive on EV/EBITDA (16.0×).** The divergence is entirely explained by leverage: ₹1,460 crore of net debt is invisible in the P/E and fully visible in the

EV. **Whenever P/E and EV/EBITDA rank two companies differently, leverage is the reason, and the EV-based measure is the more honest one.** Note also Konkan's 52% CFO/EBITDA — its profit is not converting to cash, which is probably why it has needed the debt.

- ☐ **BWL sits mid-table on quality and above the median on P/E.** It trades at a 22% P/E premium to the median on roughly median quality metrics. That is the fact relative valuation delivers. Whether it is justified requires the work in section 9.

- ☐ **Vidarbha is the genuinely interesting one.** Median quality, slowest growth, and the lowest multiple among the decent businesses. A thesis here would need to answer: is the slow growth structural or temporary?

⚠ **The three ways a peer table misleads**

One: the peers are not peers. Two companies in "auto components" where one supplies internal combustion engine parts and the other supplies electric drivetrains have opposite futures. Define the comparable set by economics, not by sector label. **Two: the whole sector is mispriced.** If every company in a hot sector trades at 60×, the "cheapest at 48×" is still expensive in absolute terms. Relative valuation is blind to this by construction. **Three: the multiples are computed inconsistently.** Trailing versus forward, standalone versus consolidated, EBITDA with or without other income, EV with or without lease liabilities. Compute all of them yourself, identically, or the table is noise.

Why a low P/E is usually low for a reason

TEXT

```
DECOMPOSING THE P/E

For a stable business, the theoretical P/E is:

   P/E  =  payout ratio x (1 + g)  /  (r - g)

   where g = sustainable growth, r = cost of equity

BWL: payout 25%, g 12.75%, r 13.0%
   P/E = 0.25 x 1.1275 / (0.130 - 0.1275)
       = 0.2819 / 0.0025
       = 112x  ???

This absurd answer is instructive. When g approaches
r, the formula explodes. It is telling you something
true: a company that can genuinely reinvest at 17%
ROE forever, while the market demands 13%, is worth
an unbounded amount. The formula fails because the
assumption fails - NOTHING grows at 12.75% forever.

Use a sustainable long-run g instead, say 6%:
   P/E = 0.25 x 1.06 / (0.13 - 0.06)
       = 0.265 / 0.07  =  3.8x

Also wrong, because a 25% payout is inconsistent with
only 6% growth - a company growing at 6% with 17% ROE
only needs to retain 35% (6/17), so it should pay out
65%:
   P/E = 0.65 x 1.06 / (0.13 - 0.06)
       = 0.689 / 0.07  =  9.8x

THE LESSON: payout, growth and ROE are NOT independent.
  g = ROE x (1 - payout)
Any P/E model where these three are set independently
is internally inconsistent. This is the most common
error in amateur valuation work.

WHAT ACTUALLY DRIVES A LOW P/E
  Low ROE          --> more capital retained per unit
                       of growth --> less cash to
                       shareholders --> lower multiple
  High risk (r)    --> higher discount --> lower multiple
  Low growth (g)   --> lower multiple
  Cyclical peak E  --> the E is temporarily inflated,
                       so the P/E is temporarily low
  Governance doubt --> the market applies a discount to
                       reported earnings it does not
                       fully believe
```

```
  Low liquidity    → small-cap illiquidity discount
  Earnings quality → CFO/PAT below 1 for years

Before concluding a stock is cheap, work through this
list and find out which of the seven applies. In the
overwhelming majority of cases, at least one does.
```

Trailing versus forward, and the cyclical inversion

Situation	Trailing P/E	Forward P/E	What is actually happening	Correct interpretation
Steady compounder	24×	21×	Earnings growing ~14%	Both are informative; forward is more relevant
Cyclical at the peak	**6×**	9×	Earnings at a cyclical maximum and beginning to fall	**A warning, not an opportunity.** Use mid-cycle earnings: average EBIT margin across a full cycle applied to current revenue.
Cyclical at the trough	**85×**	22×	Earnings near zero at the bottom of the cycle	Often the best entry point. Value the company on replacement cost or mid-cycle earnings, not trailing P/E.
One-off gain in the base	11×	19×	Last year included a large asset-sale gain	Normalise: strip out the exceptional item before computing anything
Turnaround in progress	Negative or 90×	14×	Earnings recovering from a loss	The forward number depends entirely on whether the recovery happens. Use EV/Sales and a normalised-margin scenario instead.

Serial acquirer	18×	13×	Forward estimates include a full year of acquired earnings	Check whether the acquisitions earn above the cost of capital before crediting the growth

Mid-cycle earnings for cyclicals, always

For steel, cement, chemicals, sugar, shipping, commodity paper, mining and real estate, trailing earnings are close to useless. The correct approach is to compute an average EBIT margin over a full cycle — typically seven to ten years, covering at least one peak and one trough — apply that margin to current revenue or current capacity, and value the resulting normalised earnings. A cement company earning a 28% EBITDA margin in a boom when its ten-year average is 17% is reporting an earnings figure that will not repeat. Applying any multiple at all to that number is a category error.

A reality check on multiples: what the market is implicitly assuming

TEXT

```
WHAT A P/E OF 24x ACTUALLY IMPLIES

Simplest version: the earnings yield
   1 / 24.2  =  4.13%

If BWL never grew, never reinvested, and paid out
100% of earnings, you would earn 4.13% a year
forever. Against a 10-year G-Sec at roughly 6.8%
(late 2026, illustrative - verify at rbi.org.in),
you would be paying for the privilege of taking
equity risk.

So the 24.2x multiple is a statement that earnings
will grow. How much?

Required: equity return of 13% (cost of equity).
Current earnings yield: 4.13%.
Gap that growth must close: 8.87% a year.

Rough test using the Gordon relationship:
   r  =  earnings yield x payout  +  g
   0.13 = 0.0413 x 0.25 + g
   0.13 = 0.0103 + g
   g  =  11.97%

The market is implicitly assuming BWL grows earnings
at about 12% a year INDEFINITELY. Against a
sustainable growth rate of 12.75% computed in section
7, that is internally consistent - but only if ROE
stays at 17% and the payout stays at 25% forever.

THE QUESTION YOU NOW HAVE TO ANSWER IS SPECIFIC:
"Can this business sustain a 17% return on equity for
the next fifteen to twenty years?" That is a question
about the moat, the industry structure, and the
competence of management. It is NOT a question about
the P/E ratio.

This is the correct use of relative valuation: not to
produce a target price, but to translate the market
price into a statement about the business, which you
can then agree or disagree with on the evidence.
```

✦ **The multiple is not the thesis**

"It trades at 14× while the sector trades at 22×" is an observation, not an argument. The argument has to be a reason why the gap should close: an earnings inflection that is visible and funded, a governance concern that has been resolved, a cyclical trough that is passing, a misunderstood segment that is about to be disclosed separately. Without a named catalyst or a named misperception, a discount to peers is simply the market's correct assessment that the business is worse — and it can persist for a decade.

Valuation Part 2 — Intrinsic and the DCF

A discounted cash flow model is the only valuation method derived from first principles rather than from what other people are paying. It is also extraordinarily sensitive to assumptions you cannot know. Both statements are true and must be held simultaneously. Build the DCF, then treat the output as a range, then use the reverse DCF — which is the genuinely useful tool.

The mechanics

TEXT

```
DCF STRUCTURE

  Enterprise value
    = PV of explicit-period free cash flow to the firm
    + PV of terminal value

  Equity value = EV − net debt (+ surplus investments)
  Per share    = equity value / diluted shares

FCFF, from the operating result:
    EBIT
  − tax on EBIT (EBIT x effective rate)
  = NOPAT
  + depreciation and amortisation   (non-cash)
  − capital expenditure
  − increase in net working capital
  = FCFF

Discount FCFF at WACC, because FCFF is the cash
available to BOTH lenders and shareholders. Never
discount FCFF at the cost of equity; never discount
FCFE at WACC. Mixing them is the single most common
structural error in amateur models.
```

Step 1: the cost of capital

TEXT

```
WACC FOR BHARAT WIDGETS LTD
(all inputs illustrative, as at late 2026 - verify
 the G-Sec yield at rbi.org.in before using)

COST OF EQUITY via CAPM
  Risk-free rate (10-year G-Sec)            6.80%
  Equity risk premium for India             6.50%
    (India ERP estimates from academic and
     practitioner sources have ranged roughly
     5.5% to 8% over the last decade; the choice
     is a judgement, and it matters enormously)
  Beta (5-year monthly vs NIFTY 500)         0.95
    (beta is unstable; many practitioners use a
     sector median or simply assume 1.0)

  Ke = 6.80% + 0.95 x 6.50%
     = 6.80% + 6.18%
     = 12.98%,  say  13.0%

COST OF DEBT
  Finance costs FY2025                         58
  Average gross debt (ex-leases)
    (600 + 612) / 2                           606
  Implied pre-tax cost  58 / 606            9.57%
    (cross-check against the borrowings note,
     which states rates of 8.9% to 10.4%)
  Use 9.50% pre-tax
  After-tax Kd = 9.50% x (1 - 0.2517) =     7.11%

CAPITAL STRUCTURE AT MARKET VALUES
  (never use book values here)
  Market cap  12 cr shares x Rs 480      = 5,760
  Debt (incl. leases)                      720
  Total capital                          6,480
  E / (D+E) = 5,760 / 6,480              = 88.9%
  D / (D+E) =   720 / 6,480              = 11.1%

WACC = 0.889 x 13.00% + 0.111 x 7.11%
     = 11.56% + 0.79%
     = 12.35%,  round to  12.5%

SANITY CHECKS BEFORE YOU USE THIS NUMBER
  - Is WACC above the risk-free rate? 12.5 > 6.8. Yes.
  - Is WACC below the cost of equity? 12.5 < 13.0.
    Yes - it must be, since debt is cheaper.
  - Is it plausible for an Indian mid-cap industrial?
    11-14% is the normal band. Yes.
  - Would a 1-point change in the ERP move the answer
```

```
a lot? Yes: ERP of 7.5% gives Ke of 13.9% and
WACC of 13.2%. Note this and test it in the
sensitivity table.
```

> ⚠ **Beta is the weakest input in the whole model**
>
> Beta is estimated from past price correlation with an index, it is unstable across measurement windows, and it changes with the frequency of observation. A beta of 0.95 computed on five years of monthly data might be 1.30 on two years of weekly data. Many thoughtful practitioners abandon it and instead build a cost of equity from a base rate plus explicit judgement-based premia for business risk, financial risk and liquidity. If you keep CAPM, at least test the valuation across a beta range of 0.8 to 1.3 and see how much it matters.

Step 2: forecast free cash flow

TEXT

```
BWL FCFF FORECAST, FY2026 to FY2030 (Rs crore)
Every assumption is stated and must be defensible.

                      FY26   FY27   FY28   FY29   FY30
Revenue growth        12.0%  11.0%  10.0%   9.0%   8.0%
  (fading towards nominal GDP; BWL's 9-year
   historical CAGR was 10.1% and the last 3 years
   averaged 11.3%, so this fade is mildly optimistic
   at the front and conservative at the back)
Revenue              2,688  2,984  3,282  3,578  3,864

EBIT margin           15.7%  15.9%  16.1%  16.3%  16.5%
  (continuing the observed trend of ~+40bps a year,
   decelerating. If you cannot name the SOURCE of
   margin expansion, assume it is flat. Margin
   expansion assumptions are where DCFs go wrong.)
EBIT                   422    474    528    583    638

Tax rate              25.17% on EBIT throughout
NOPAT                  316    355    395    436    477

D&A as % of revenue    5.5%   5.5%   5.5%   5.5%   5.5%
D&A                    148    164    181    197    213

Capex as % of revenue  9.0%   8.5%   8.0%   7.5%   7.0%
  (declining as the current expansion completes;
   FY2025 actual was 11.0% of revenue)
Capex                 (242)  (254)  (263)  (268)  (271)

Increase in NWC at 18% of incremental revenue
  (FY2025 actual was 10.7%; using 18% is deliberately
   conservative because working-capital efficiency
   gains eventually stop)
Increase in NWC        (52)   (53)   (54)   (53)   (52)
                      _____  _____  _____  _____  _____
FCFF                   170    212    259    312    367

DISCOUNTING AT WACC 12.5%
Discount factor      0.8889 0.7901 0.7023 0.6243 0.5549
PV of FCFF             151    168    182    195    204

SUM OF PV OF EXPLICIT FCFF          =  Rs 900 cr
```

Step 3: terminal value, computed both ways

TEXT

```
TERMINAL VALUE - METHOD 1: PERPETUITY GROWTH

  TV = FCFF(FY30) x (1 + g) / (WACC - g)

  Choosing g: terminal growth must not exceed the
  long-run nominal growth rate of the economy, or
  the company eventually becomes larger than India.
  Long-run Indian nominal GDP growth of roughly
  5-6% real plus 4-5% inflation suggests 9-11% is
  the theoretical ceiling; most practitioners use
  4-6% to stay conservative. Use g = 5.0%.

  TV = 367 x 1.05 / (0.125 - 0.050)
     = 385.4 / 0.075
     = Rs 5,139 cr

  PV of TV = 5,139 x 0.5549 = Rs 2,852 cr

  TV as a share of total EV = 2,852 / 3,752 = 76%
  *** THREE QUARTERS OF THE VALUE IS IN A NUMBER
      THAT DEPENDS ON TWO ASSUMPTIONS ABOUT THE
      YEAR 2030 AND BEYOND. ***

TERMINAL VALUE - METHOD 2: EXIT MULTIPLE

  FY2030 EBITDA = EBIT 638 + D&A 213 = 851
  Apply an exit EV/EBITDA multiple.

  At 9.0x:  TV = 7,659  ->  PV = 4,250
  At 11.0x: TV = 9,361  ->  PV = 5,195

  THE TWO METHODS DISAGREE VIOLENTLY. Why?

  Back out the multiple implied by the perpetuity
  method:  5,139 / 851 = 6.0x EV/EBITDA

  So a 5% perpetuity growth rate implies BWL will
  trade at 6.0x EBITDA in FY2030, against 12.6x
  today. The perpetuity method is saying the
  multiple MUST compress - and it is right to,
  because in the forecast capex (271) still exceeds
  D&A (213), so the business consumes cash to stand
  still, and a business like that is worth less per
  rupee of EBITDA than one that does not.

  USE BOTH, ALWAYS, AND RECONCILE THEM. If the
  implied exit multiple from your perpetuity method
  is wildly different from where the company and its
```

```
  peers trade today, one of your assumptions is
  wrong and you have just found it. This
  cross-check is the most valuable five minutes in
  DCF work.
```

Step 4: bridge to value per share

TEXT

```
BWL DCF - BASE CASE (WACC 12.5%, g 5.0%)

  PV of explicit FCFF, FY26-FY30          900
  PV of terminal value                  2,852
                                        -----
  ENTERPRISE VALUE                      3,752

  Less: gross debt (incl. lease
        liabilities 120)                 (720)
  Add:  cash and bank balances            150
  Add:  non-current investments at
        a conservative book value          60
  Less: minority interest                   0
  Less: contingent liabilities judged
        probable (judgement; none here)     0
                                        -----
  EQUITY VALUE                          3,242

  Diluted shares                     12.1 crore
  VALUE PER SHARE                  Rs 267.93
                                   say Rs 268

  CURRENT MARKET PRICE             Rs 480
  IMPLIED OVER-VALUATION                79%

  DO NOT STOP HERE. A single-point DCF output is
  worth very little. Go to the sensitivity table.
```

Step 5: the sensitivity table, which is the real output

Value per share (₹)	g = 4.0%	g = 4.5%	g = 5.0%	g = 5.5%	g
WACC 11.5%	276	295	317	342	37
WACC 12.0%	254	271	289	311	33
WACC 12.5%	235	250	**268**	284	30
WACC 13.0%	219	231	245	260	27
WACC 13.5%	204	214	226	240	25

Read that table honestly

Two assumptions that nobody can know — a discount rate within a plausible two-point band, and a perpetual growth rate within a plausible two-point band — produce a valuation range of **₹204 to ₹372, an 82% spread**. And that is *before* varying revenue growth, margins or capex, each of which moves the answer further. Anyone who quotes a DCF target price to the rupee without showing you this table either has not built one or is not being straight with you. The honest output of a DCF is a range and a set of named dependencies, not a number.

The three-scenario DCF

Assumption	Bear	Base	Bull
Revenue CAGR FY26–30	5.0%	10.0%	14.0%
FY2030 EBIT margin	13.0%	16.5%	19.0%
Capex as % of revenue by FY30	8.5%	7.0%	6.5%
Incremental NWC / incremental revenue	25%	18%	12%
WACC	13.5%	12.5%	12.0%
Terminal growth	4.0%	5.0%	5.5%
Value per share	**₹148**	**₹268**	**₹421**
Probability	25%	50%	25%
Probability-weighted value	**0.25×148 + 0.50×268 + 0.25×421 = 37 + 134 + 105 = ₹276**		

At ₹480, even the bull case does not justify the price. That is a clean, actionable conclusion — and it is a conclusion about the price, not about the company. BWL may be an excellent business. The analysis says you are being asked to pay more than the cash flows plausibly support.

Reverse DCF: the tool you should actually use

Instead of guessing the future and deriving a price, take the price as given and derive what future it implies. You then only have to judge whether that future is plausible — a much easier question than forecasting it yourself.

TEXT

```
REVERSE DCF FOR BWL AT Rs 480

  Market cap             12.1 cr x 480 = Rs 5,808 cr
  Add net debt (720 - 150 - 60)        = Rs   510 cr
  IMPLIED ENTERPRISE VALUE             = Rs 6,318 cr

  Hold WACC at 12.5% and terminal growth at 5.0%.
  Solve for the revenue growth and margin path that
  produces an EV of 6,318 instead of 3,752.

  Iterating, one combination that works:

    Revenue CAGR FY26-FY35 (10 years)      16.5%
    EBIT margin rising 15.5% → 20.0%
      by FY2033 and held
    Capex falling to 6.5% of revenue
    Incremental NWC at 15% of incremental revenue
    Terminal growth 5.0%, WACC 12.5%

  NOW JUDGE THE ASSUMPTIONS, NOT THE MODEL:

  Q1. Is a 16.5% revenue CAGR for TEN years
      plausible?
      Revenue would go from Rs 2,400 cr to Rs 11,110
      cr. BWL's best 3-year run was 11.3%. Its
      9-year CAGR was 10.1%. The addressable market
      would need to grow substantially AND BWL would
      need to gain share throughout.
      → Demanding. Possible, not probable.

  Q2. Is a 20% EBIT margin plausible?
      The best peer (Deccan) earns ~20% EBIT. BWL
      would have to match the sector's highest-
      quality operator.
      → Demanding.

  Q3. Can capex fall to 6.5% of revenue WHILE
      growing revenue at 16.5%?
      Growing 16.5% a year with capex at 6.5% of
      revenue requires capital productivity
      materially better than the company has ever
      achieved.
      → The three assumptions are in tension with
         each other.

  CONCLUSION: at Rs 480 the market is pricing BWL to
  simultaneously grow faster than it ever has,
  achieve best-in-class margins, and become more
```

```
capital-efficient while doing so. Each is
individually possible. All three together, for a
decade, is a demanding bet.

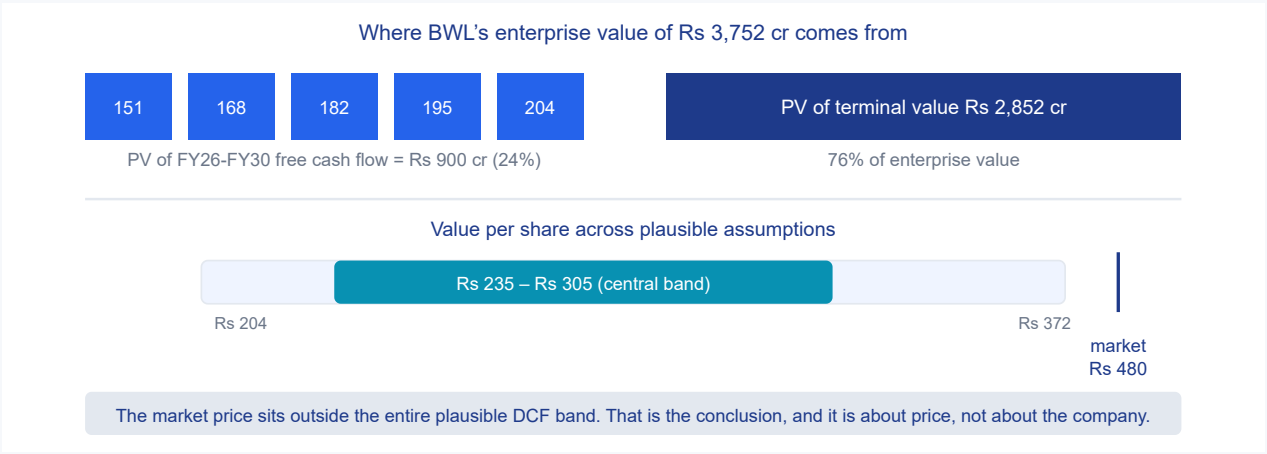
THAT is a usable conclusion. It does not require
you to forecast anything. It requires you to
judge a specific, stated set of claims.
```

Use the reverse DCF as a screening filter

Run a reverse DCF on anything you are tempted to buy, then write down the implied growth and margin assumptions in one sentence. "At this price the market needs 18% growth for ten years and margins at an all-time high" is a sentence you can test against the annual report, the industry data and the concall. "My DCF says it is worth ₹620" is a sentence you cannot test against anything. The reverse DCF converts an unanswerable forecasting question into an answerable plausibility question, which is the only realistic edge an individual investor has in valuation.

Where DCFs break, honestly

Fragility	Magnitude	Mitigation
Terminal value dominates	Typically 60–80% of total EV. For a high-growth company it can exceed 90%.	Extend the explicit forecast to 10–15 years so the terminal value falls to 40–55%. Cross-check the implied exit multiple.
Discount rate is a guess	A 1-point change in WACC moves value by roughly 15–25%	Always show a sensitivity table. Treat the output as a band.
Margin expansion is assumed	Assuming margins rise 300 bps over five years can add 25–35% to the value	**Default to flat margins unless you can name the specific mechanism.** This single discipline removes most DCF over-valuation.
Capex is underestimated	Models routinely assume capex converges to depreciation; real businesses rarely do	Use the ten-year historical ratio of capex to revenue as a floor
Working capital is ignored	For a growing business this can be the difference between positive and negative FCF	Model incremental NWC as a percentage of incremental revenue, using the ten-year historical ratio
Competition is absent	Models implicitly assume returns above the cost of capital persist forever. Economics says they should not.	Fade ROIC towards WACC in the terminal period, or shorten the competitive-advantage period
Dilution is ignored	ESOPs, warrants and QIPs transfer value away from existing shareholders	Use fully diluted share count and model expected future issuance
Anchoring on the current price	The most insidious. Analysts adjust assumptions until the output approximates the market price.	Write down every assumption *before* computing the output, and keep the file. If you change an assumption afterwards, record why in writing.



Three quarters of a typical DCF's value sits in the terminal value, and the plausible output is a wide band rather than a point.

Qualitative Analysis: Moats, Industry and Management

The numbers tell you what happened. Qualitative analysis tells you whether it will keep happening. A high return on capital is an invitation to competitors; the only reason it persists is that something stops them. Identifying that something — or establishing that there is nothing — is the part of analysis that cannot be screened for and therefore the part where an individual can still add value.

The five sources of durable competitive advantage

Moat	Mechanism	Financial signature	Indian context	How it erodes
Intangible assets: brand	Customers pay more for an identical product because of trust, habit or status	Gross margin well above sector; advertising spend sustained at 5–12% of revenue; pricing increases absorbed without volume loss	Strongest in packaged foods, personal care, paints, and certain consumer durables, where distribution depth and decades of recall reinforce each other	Changing consumer preferences; direct-to-consumer entrants with lower customer-acquisition costs; retailer private labels
Intangible assets: regulatory	Licences, approvals or patents legally restrict entry	Very high and stable ROCE; capex lumpy and tied to approvals	Pharmaceutical approvals, banking licences, telecom spectrum, port and airport concessions, tobacco, liquor retail in certain states	Patent expiry; policy change; licence renewal on worse terms. **A regulatory moat is a political asset and can be legislated away.**
Switching costs	Changing supplier costs the customer money, time, risk or retraining	Very high customer retention; long contracts; high revenue from existing customers; receivable days driven by the customer's process, not credit risk	Enterprise software, core banking systems, specialty chemicals qualified into a customer's product, auto components designed into a platform, medical devices	A generational technology change that forces every customer to switch anyway
Network effects	The product becomes more valuable as more people use it	Winner-take-most market shares; marketing spend falling as a	Exchanges and depositories, payment networks,	Multi-homing (users on several platforms at once); regulatory

		share of revenue once scale is reached	classifieds and marketplaces, some B2B platforms	intervention; a subsidised entrant buying scale
Cost advantage	Structurally lower cost per unit: scale, process, location, or access to a resource	EBITDA margin above peers with similar gross margin; the ability to stay profitable at prices that put competitors into losses	Captive limestone or coal reserves, a plant adjacent to a port, a distribution network too dense to replicate economically, genuine manufacturing scale	A competitor builds equivalent scale; the input-cost advantage normalises; technology changes the cost curve

The test that separates a moat from a good year

A moat should be visible in the **persistence** of returns, not their level. Look at ten years of ROCE. A company whose ROCE stayed between 22% and 28% through a recession, a commodity spike and a competitive entry has something structural. A company whose ROCE went 9%, 14%, 31%, 24%, 11% is a cyclical, whatever the story says. **Low variance in ROCE at a high level is the financial fingerprint of a moat**, and it is one of the few qualitative judgements you can make quantitatively.

Industry structure: Porter's five forces, applied

Force	Question	Evidence to look for in Indian filings	Favourable
Threat of new entry	How hard is it to start competing?	Capital required for minimum efficient scale; licences; the number of new entrants in the last decade (visible in industry association data and IPO filings)	High capital requirement, long approval cycles, established distribution that takes years to build
Buyer power	Can customers dictate price?	Customer concentration disclosed in the risk factors and segment note; receivable days; whether price revisions require customer agreement	Fragmented customers, each small relative to the supplier. Dangerous: one OEM accounting for 40% of revenue.
Supplier power	Can suppliers dictate cost?	Raw-material concentration; whether gross margin moves with a commodity index; related-party supply arrangements	Multiple substitutable suppliers; the ability to pass through input costs within a quarter
Threat of substitutes	Can the customer solve the problem a different way?	Technology shifts, regulatory mandates, changing end-use patterns	No economically viable substitute within the forecast horizon
Rivalry	How brutally do incumbents compete?	Industry-wide margin trend; the share of the market held by the top five; whether capacity additions by any player trigger price wars	Consolidated industry with rational pricing behaviour. Dangerous: fragmented industry with excess capacity and undifferentiated product, which describes much of Indian commodity manufacturing.

TAM, runway, and why most TAM numbers are useless

Total addressable market figures in investor presentations are almost always sourced from a commissioned consultancy report and are almost always too large, because they count the entire category rather than the segment the company can realistically serve at its price point. Three disciplines fix this.

1. **Build TAM bottom-up, not top-down**
   Top-down: "the Indian widget market is ₹80,000 crore". Bottom-up: number of plausible customers × units each buys per year × realistic price. The two answers typically differ by a factor of three to ten, and the bottom-up one is usually right.

2. **Compute the implied share, then sanity-check it**
   If a ₹2,400 crore company is to grow at 16% for ten years, it reaches ₹10,600 crore. Against a credible serviceable market of ₹30,000 crore growing at 8%, that is a move from roughly 8% share to 16% share. Ask who loses that share, and whether they will let it happen quietly.

3. **Convert runway into years**
   The useful question is not "how big is the market" but "at the current growth rate, how many years before the market constrains the company?" If the answer is under five years, the terminal assumptions in your valuation are doing all the work and the growth story is already mostly priced.

Capital allocation: the most important thing management does

TEXT

CAPITAL ALLOCATION SCORECARD – BWL, FY2016 to FY2025
(cumulative, Rs crore)

SOURCES OF CAPITAL		
Cumulative CFO	2,614	
Net debt raised	148	
Equity raised	0	
TOTAL AVAILABLE	2,762	
USES OF CAPITAL		
Capex (organic)	1,682	61%
Acquisitions	310	11%
Dividends	426	15%
Buybacks	0	0%
Debt reduction	200	7%
Increase in cash and investments	144	5%
	2,762	

THE FIVE QUESTIONS THAT SCORE IT

1. Did organic capex earn above the cost of capital?
   Incremental ROCE over the period was ~30% pre-tax (~22% post-tax) vs a WACC of 12.5%. PASS.
2. Did acquisitions earn above the cost of capital?
   The one acquisition is earning ~8.5% post-tax on the price paid, vs a 12.5% WACC. FAIL.
3. Was the dividend affordable?
   Cumulative dividend 426 vs cumulative FCF (2,614 – 1,682) = 932. Payout of 46% of FCF. PASS.
4. Was there dilution?
   Share count 11.4 → 12.0 crore, +5.3% over nine years (0.6% a year), all from ESOP exercise. Acceptable. PASS.
5. Was debt used sensibly?
   Net debt / EBITDA fell from 2.9x to 0.89x while the business grew. PASS.

SCORE: 4 of 5. The weak point is acquisitions, which is also the most common weak point in Indian corporate capital allocation. Watch whether management does more of them – a single value-destroying acquisition is a mistake, a pattern of them is a strategy, and the strategy should change your valuation.

Promoter quality, holding and pledges

Signal	Where to find it	Benign reading	Concerning reading
Promoter holding level	Quarterly shareholding pattern	Stable or rising, typically 40–70%. Skin in the game aligned with minorities.	Falling steadily without explanation. Very high (above 75%, the SEBI minimum public shareholding limit) with poor liquidity. Very low (under 20%) with no other anchor.
Pledged shares	Same filing, a separate column	Zero, or a small amount with a stated purpose and a declining trend	**Any material pledge is a risk amplifier.** If the price falls, the lender makes a margin call; if the promoter cannot meet it, shares are sold into a falling market, which pushes the price down further and triggers more selling. A reflexive loop that has repeatedly taken Indian mid-caps down 70–90% in weeks. Rising pledge percentage is a serious flag.
Related-party transactions	Notes to accounts; also the LODR disclosure of material RPTs requiring shareholder approval	Small, at arm's length, declining as a share of revenue, with a clear commercial rationale	Material sales to or purchases from promoter-owned entities; loans and advances to group companies; royalty or brand fees paid to a promoter entity; guarantees given for group debt. **This is the single most common mechanism by which value leaves a listed Indian company.**
Remuneration	Corporate governance report; Section 197 disclosures	Reasonable relative to PAT and to peers, with a variable component linked to disclosed targets	Promoter-director remuneration growing much faster than profit, or a large commission on profit in a year of weak cash generation
Board independence	Corporate governance report	Genuinely independent directors with relevant	Independent directors who are family friends, hold many other directorships, or attend rarely.

		expertise, regular attendance, and chairs of audit and nomination committees who are independent	Rapid independent-director turnover.
Auditor	Auditor's report; exchange announcements	A long-standing, reputable firm; rotation at the statutory interval under Section 139	**An auditor resigning mid-term is one of the strongest negative signals in Indian markets.** Read the resignation letter, which must be disclosed under LODR, and read what the auditor says it could not satisfy itself about.
Past conduct	SEBI orders, exchange surveillance actions, the promoter history in any group DRHP	Clean	Prior SEBI enforcement, defaults at other group entities, a history of failed ventures left with creditors

✕ **Governance is binary, not a discount**

Many investors treat weak governance as something to be compensated for by a lower price — "it is cheap, so the governance risk is in the price". The arithmetic does not work. If the promoter is extracting value through related-party transactions, the extraction scales with the business and is not a one-off. If accounts are misstated, your entire analysis rests on fiction and the "cheapness" is an illusion. In practice the correct response to material governance concerns is to decline the investment entirely, not to size it smaller. The opportunity cost of skipping a cheap company with bad governance is one missed idea. The cost of being wrong is total.

Accounting Red Flags and Forensic Basics

Forensic analysis is not about proving fraud. It is about identifying companies where the published numbers are internally inconsistent, and then declining to invest rather than investigating further. You are not an auditor, you have no subpoena power, and the asymmetry is in your favour: there are thousands of listed companies and you only need a handful. **The correct response to an unresolved red flag is to move on.**

⚠ **On naming companies**

Every pattern described below has appeared in Indian corporate failures, and each is described generically because this guide makes no allegation against any identifiable company. Where you want case studies, read SEBI adjudication orders, NCLT insolvency judgements and forensic audit reports commissioned by lenders — all are public and far more rigorous than press coverage.

The red flag catalogue, grouped by where it shows up

#	Flag	Where to look	What it suggests	Severity
1	Cumulative CFO materially below cumulative PAT over 5–10 years	Cash flow statements	Earnings not converting to cash by any mechanism	Severe
2	Receivable days rising for three or more consecutive years	Balance sheet; ageing schedule	Channel stuffing, uncollectible sales, or related-party round-tripping	Severe
3	Receivables over one year old rising, with a falling provision percentage	Schedule III ageing table	Losses being deferred	Severe
4	Inventory growing faster than revenue for several years, concentrated in finished goods	Inventory note	Demand has fallen, or costs are being parked in inventory	Material
5	CWIP large and not converting to PPE for multiple years	PPE note; CWIP ageing schedule	Stalled project, cost overrun, or capitalised operating expenses	Severe
6	Capex consistently 2.5× depreciation with no capacity or revenue growth	Cash flow statement vs capacity disclosures	Capital leaving the company without producing assets	Severe
7	Depreciation as a percentage of gross block falling	PPE note over ten years	Useful lives extended to raise reported profit	Material

8	Large and growing "intangibles under development"	Intangibles note	Development spend capitalised rather than expensed	Material
9	Exceptional items in more than three of ten years	P&L	Recurring costs presented as one-offs	Material
10	Other income exceeding 25–30% of PBT	P&L and its note	Reported profit is not operating profit	Material
11	Related-party revenue, purchases or receivables rising as a share of total	Related-party note	Value extraction or fabricated transactions	Severe
12	Loans, advances or guarantees to promoter group entities	Related-party note; Schedule III disclosures; contingent liabilities	The listed company funding or underwriting private interests	Severe
13	Promoter pledge rising, especially above 25% of promoter holding	Shareholding pattern	Promoter leverage; a reflexive downside loop in any drawdown	Severe
14	Auditor resignation mid-term, or a change away from a reputable firm	Exchange announcements; auditor's report	Disagreement over accounting treatment the auditor could not resolve	Severe
15	Qualified opinion, adverse opinion, disclaimer, or an emphasis-of-matter on going concern	Auditor's report	The auditor is publicly disclaiming responsibility for part of the accounts	Severe

16	CFO resignation, or repeated CFO turnover	Exchange announcements	The person who signs the numbers is unwilling to continue	Material
17	Large cash balance held alongside expensive debt	Balance sheet; borrowings note	Either poor treasury management, or cash that is not genuinely available or not genuinely there	Material
18	Subsidiaries with significant assets but minimal disclosure, especially offshore	Consolidated accounts; subsidiary statement (Form AOC-1)	Activity structured out of sight	Material
19	Contingent liabilities large relative to equity, especially guarantees for group entities	Contingent liabilities note	Off-balance-sheet obligations that can become real	Material
20	Frequent equity raising despite reported profitability	Cash flow financing section; share capital movements	Profits are not cash, or capital is leaking somewhere	Severe
21	Effective tax rate far below the statutory rate for years, without a clear exemption	Tax reconciliation note	Either an expiring benefit that will reverse, or book profits that are not taxable because they are not real	Material

22	A sudden change in accounting policy, estimate, or the financial year-end	Notes; significant accounting policies	Comparability deliberately broken in a year when comparison would be unflattering	Material
23	Revenue heavily concentrated in Q4 relative to Q1–Q3	Quarterly filings	Year-end revenue recognition pressure	Material
24	Exchange surveillance action: ASM, GSM, or a long-term additional surveillance framework	NSE/BSE surveillance lists	Unusual price or volume behaviour the exchange has flagged	Material
25	Resignation of independent directors citing "personal reasons" shortly before results	Exchange announcements	Boardroom disagreement. LODR requires the resignation letter to be disclosed — read it.	Severe

Three cross-checks that catch most problems in ten minutes

TEXT

```
CROSS-CHECK 1: THE CASH TRIANGLE
  Over ten years, does
    cumulative PAT
    ~= cumulative CFO  (allowing for depreciation
                        add-back, so CFO should be
                        HIGHER)
    and is cumulative FCF positive?
  If PAT is strongly positive and cumulative FCF is
  strongly negative with no completed capex cycle to
  show for it, stop.

CROSS-CHECK 2: THE GROWTH TRIANGLE
  Over ten years, compare the growth rates of:
    Revenue
    Receivables
    Inventory
  If receivables or inventory are compounding
  materially faster than revenue, the balance sheet
  is absorbing the growth. Quantify:
    Revenue CAGR          12%
    Receivables CAGR      22%
    → receivables as % of revenue roughly DOUBLES
       over 8 years. Where is the cash?

CROSS-CHECK 3: THE ASSET TRIANGLE
  Compare the growth rates of:
    Gross block (PPE at cost)
    Capacity (tonnes, units, square feet)
    Revenue
  If gross block grows 15% a year and capacity grows
  4% a year, the company is spending far more per
  unit of capacity than it used to. Either input
  costs for plant have risen sharply (check against
  a peer building similar capacity), or the capex
  number includes things that are not plant.

  Peer benchmarking is powerful here: if a competitor
  builds a 1 million tonne plant for Rs 900 crore and
  your company reports Rs 1,800 crore for the same
  capacity, you have a specific, answerable question
  to put to the company at the AGM.
```

The Beneish M-score and Altman Z-score: useful, limited

Two published statistical screens exist. The **Beneish M-score** combines eight accounting ratios into a single earnings-manipulation probability; the **Altman Z-score** combines five into a bankruptcy-risk measure. Both were calibrated on US data from earlier decades and neither transfers cleanly to Indian accounting or to Indian capital structures — and the Z-score in particular is meaningless for financial companies. Use them as a ranking device across a screened list to decide where to look first, never as a verdict. The three cross-checks above, done by hand on the actual filings, are more reliable than either score.

Sector-Specific Analysis

Almost everything in sections 3 to 9 assumes a company that buys inputs, makes something, and sells it. For several large parts of the Indian market that model does not apply at all, and applying the standard ratios produces confident nonsense. Banks have no revenue line in the ordinary sense; their inventory is money and their biggest cost is a judgement about future losses. This section gives the correct framework for each major sector.

Banks and NBFCs: a completely different toolkit

Ratios that are meaningless for a bank

Debt-to-equity (a bank is leveraged 8–12× by design — deposits are its raw material, not a sign of distress). EV/EBITDA (interest is operating revenue, so EBITDA has no meaning). Current ratio (the maturity mismatch between short deposits and long loans *is* the business). Inventory and receivable days. Free cash flow as ordinarily computed. **Using any of these on a bank will produce a conclusion that is not merely wrong but backwards.**

Metric	Definition	What it tells you	Rough reference range
Net interest margin (NIM)	(Interest earned − interest expended) ÷ average interest-earning assets	The spread the bank captures between lending and funding. The core profitability driver.	Indian private banks commonly 3.0–4.5%; public sector banks lower; microfinance and consumer NBFCs much higher, with correspondingly higher credit costs
CASA ratio	(Current + savings account deposits) ÷ total deposits	The proportion of funding that is cheap and sticky. The single biggest structural advantage a bank can have.	A high CASA ratio is a durable franchise asset; it is built over decades through branch networks and salary accounts
GNPA %	Gross non-performing assets ÷ gross advances	The share of the loan book that has stopped performing, per RBI's 90-day rule	Compare against the bank's own history and against peers in the same lending mix
NNPA %	(GNPA − provisions held) ÷ net advances	Bad loans not yet provided for. **This is the number that eventually hits equity.**	Low single digits or below is the usual comfort threshold; the gap between GNPA and NNPA is the provisioning buffer
Provision coverage ratio (PCR)	Provisions held ÷ GNPA	How much of the recognised bad book is already absorbed	Higher is more conservative. A falling PCR alongside flat GNPA means losses are being deferred.
Credit cost	Provisions for the year ÷ average advances	The annual cost of lending mistakes. Deduct it mentally from NIM to get the true spread.	Varies enormously by segment: secured corporate lending is low, unsecured consumer and microfinance is several times higher

Capital adequacy (CRAR)	Regulatory capital ÷ risk-weighted assets, per Basel III as applied by RBI	Loss-absorbing capacity and the headroom to grow the book. A bank below the regulatory minimum cannot grow and may need to raise equity at a bad time.	RBI prescribes minimums including buffers; check the current requirement at rbi.org.in
CET-1 ratio	Common equity tier 1 ÷ RWA	The highest-quality capital. The number that matters in a crisis.	Compare against the regulatory minimum plus the capital conservation buffer
Cost-to-income	Operating expenses ÷ (net interest income + other income)	Operating efficiency	Lower is better; digital-first lenders should structurally beat branch-heavy ones
ROA	PAT ÷ average total assets	**The right profitability measure for a bank**, because ROE can be manufactured with leverage	Around 1.5%+ is strong for an Indian bank; below 0.5% indicates a problem
ROE	PAT ÷ average equity	Useful only alongside ROA and leverage — ROE = ROA × leverage	—
P/B and P/ABV	Price ÷ book value, and price ÷ adjusted book value (book value less net NPA)	**The primary valuation multiple for banks.** Adjusted book value is the more honest version because it removes bad loans not yet written off.	A bank earning a sustainably high ROE deserves a higher multiple of book; one earning below its cost of equity should trade below book
SMA 0/1/2 and restructured book	Special-mention accounts overdue 1–30, 31–60 and 61–90 days; plus restructured advances	**Leading indicators.** Today's SMA-2 is next quarter's NPA.	Disclosed in investor presentations; chronically under-read

⚠ **NBFCs differ from banks in the way that matters most**

An NBFC cannot take demand deposits, so it funds itself from banks, bonds, commercial paper and securitisation. That makes it vulnerable to a funding freeze in a way a deposit-funded bank is not: when wholesale markets close, an NBFC with short-term borrowings funding long-term loans faces an asset-liability mismatch it cannot refinance. Always read the **ALM (asset-liability maturity) statement** in the annual report, which buckets assets and liabilities by maturity. Cumulative negative gaps in the short buckets are the vulnerability. Also check the funding mix: heavy reliance on commercial paper is cheap until it is unavailable.

IT services

What to track	Why
Constant-currency revenue growth	Strips out the rupee's movement. Reported rupee growth flatters in a depreciating-rupee year and nothing operational has changed.
Utilisation (including and excluding trainees)	Billed hours as a share of available hours. The primary short-term margin lever. High utilisation means little slack for a new deal.
Attrition (LTM voluntary)	Replacement and retraining costs; high attrition damages delivery quality and client relationships. A lead indicator of margin pressure.
Offshore / onsite mix	Offshore work carries far higher margins. A shift onsite compresses margins even with flat revenue.
Client metrics	Number of clients above the ₹100 crore, ₹50 crore and ₹10 crore annual revenue thresholds; revenue from the top 5 and top 10 clients. Mining existing accounts is cheaper than winning new ones.
Total contract value and deal wins	The forward book. Distinguish net-new TCV from renewals — companies often blur the two.
Revenue per employee	The clearest measure of whether the business is moving up the value chain or remaining a labour-arbitrage model.
Pyramid and fresher hiring	A healthier pyramid (more juniors per senior) lowers cost. Fresher hiring volumes signal management's demand outlook 6–12 months ahead.

FMCG and consumer

- ☐ **Volume growth versus value growth.** Value growth of 12% composed of 2% volume and 10% price is a very different business from 10% volume and 2% price. Only volume growth is

unambiguously good; price-led growth in a high-inflation year is just inflation.

- [ ] **Advertising and promotion as a percentage of sales.** Cutting A&P boosts this year's margin and erodes the brand. Watch for margin expansion that coincides with falling A&P.

- [ ] **Distribution reach** — direct and total outlets covered, rural versus urban split, and the growth of modern trade, e-commerce and quick commerce as channels. Channel shift changes both margin and working capital.

- [ ] **Gross margin versus input costs.** Palm oil, crude derivatives, milk, wheat and packaging move together; a company that holds gross margin through an input spike has pricing power.

- [ ] **New product contribution** — the share of revenue from products launched in the last three years. Innovation keeps a portfolio relevant.

- [ ] **Negative working capital** is the norm; if it turns positive, the trade is pushing back.

Pharmaceuticals

- [ ] **Geography mix** — US generics (high margin, brutal price erosion), Europe (tender-driven), India branded (stable, high margin, relationship-driven), emerging markets, API and contract manufacturing. Each has different economics; the consolidated margin is a blend you must disaggregate.

- [ ] **US price erosion** — a structural single-to-double-digit annual decline in generic prices. A company must launch new products merely to stand still.

- [ ] **ANDA and DMF filings, approvals and the pending pipeline** — the forward revenue engine. Also track Para IV filings and first-to-file exclusivities, which are high-margin but temporary.

- [ ] **Regulatory inspection status** — USFDA Form 483 observations, warning letters and import alerts on specific plants. A warning letter on a plant contributing 30% of US revenue is a material, quantifiable event. Check the inspection classification for each facility.

- [ ] **R&D as a percentage of sales**, and critically whether it is expensed or capitalised. Conservative companies expense it.

- [ ] **Working capital is structurally heavy** — 100–180 days is normal, driven by long manufacturing and regulatory cycles.

Cyclicals and commodities

TEXT

```
VALUING A CYCLICAL - THE MID-CYCLE METHOD

Never use trailing earnings. Instead:

STEP 1  Take 10 years of EBITDA margin, covering at
        least one peak and one trough.
          e.g. 8%, 12%, 19%, 24%, 21%, 11%, 6%,
               9%, 17%, 23%   →  average 15%

STEP 2  Apply the mid-cycle margin to CURRENT
        revenue or, better, to current capacity at
        a mid-cycle realisation.
          Current revenue  Rs 6,800 cr
          Mid-cycle EBITDA 6,800 x 15% = Rs 1,020 cr

STEP 3  Apply a mid-cycle multiple (NOT the peak
        multiple), typically 6-9x EV/EBITDA for an
        Indian commodity producer.
          EV = 1,020 x 7 = Rs 7,140 cr
          Less net debt  = Rs 2,400 cr
          Equity value   = Rs 4,740 cr

STEP 4  Cross-check against REPLACEMENT COST.
        What would it cost to build this capacity
        today? A cyclical trading far below
        replacement cost is usually near a trough;
        far above it, new capacity will be built
        and the cycle will turn.

THE P/E INVERSION, STATED PLAINLY
  At the peak:   E is at a maximum  → P/E looks LOW
                 → the stock looks cheap
                 → it is the most dangerous moment
  At the trough: E is near zero     → P/E looks HUGE
                 → the stock looks expensive
                 → it is often the best entry

  For cyclicals, P/B and EV per tonne of capacity
  are more stable anchors than P/E.

WHAT ELSE TO TRACK
  - Global and domestic capacity utilisation
  - Announced capacity additions across the industry
    (the single best predictor of the next downturn)
  - Spreads: the gap between output and input prices
    (refining margins, steel spreads, chemical
    delta), which is the real driver, not the
    output price alone
```

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  - Import parity prices and any anti-dumping duties
  - Inventory levels across the supply chain
  - Cost-curve position: a producer in the bottom
    quartile of the global cost curve survives a
    downturn that kills the top quartile
```

Other sectors, briefly

Sector	Key metrics	Main trap
Real estate	Pre-sales (bookings) value and volume, collections, net debt, unsold inventory in months, land bank at cost versus market, RERA registrations and project completion record	Revenue recognition timing bears almost no relation to cash. Use **pre-sales and collections**, not revenue. Land valued at decades-old cost makes book value meaningless. Leverage plus a demand downturn is the classic failure mode.
Utilities and power	Plant load factor, regulated return on equity, receivable days from state distribution companies, fuel supply agreements, PPA tenure and tariff	Returns are often regulated, so the upside is capped while the downside from receivable stress at state discoms is not. Check the ageing of government receivables.
Telecom	ARPU, subscriber additions, data usage per subscriber, churn, spectrum liabilities and payment schedule, capex intensity, network coverage	Enormous capex and spectrum obligations sit outside EBITDA. Use EV/EBITDA with all spectrum liabilities included in debt.
Insurance	Life: annualised premium equivalent, value of new business, VNB margin, embedded value, persistency at 13/25/37/49/61 months. General: combined ratio, loss ratio, expense ratio, solvency ratio	Standard accounting profit is close to meaningless. Embedded value and VNB are the correct framework for life insurers, and a combined ratio above 100% means underwriting losses.
Hotels and aviation	Occupancy, average room rate, RevPAR; for airlines: load factor, RASK, CASK, yield, fleet and fuel hedges	Extreme operating leverage combined with high financial leverage. Both industries have destroyed vast amounts of capital globally over full cycles.
Capital goods and EPC	Order book, order inflow, book-to-bill ratio, execution cycle, retention money, claims receivable	Order book quality matters more than size: a large book at thin margins with slow-paying government clients is a liability. Unbilled revenue and receivables tell the real story.

Macro and Top-Down Inputs

Macro forecasting is close to impossible and most investors would do better ignoring it entirely. What is useful is understanding the **transmission mechanisms**: how a change in interest rates, inflation, the currency or policy actually reaches a particular company's cash flows. You will not predict the repo rate. You can know, in advance, what a 200 basis point rate rise would do to the company you own.

Interest rates and valuations

TEXT

```
WHY RATES MATTER SO MUCH TO VALUATION

The discount rate r in every valuation equation
contains the risk-free rate. Move the risk-free rate
and you move every asset price.

BWL, recomputed at different G-Sec yields
(ERP held at 6.5%, beta 0.95, g at 5.0%)

  G-Sec   Ke      WACC    DCF value/share   Change
  5.80%   11.98%  11.4%      Rs 322         +20%
  6.80%   12.98%  12.5%      Rs 268          base
  7.80%   13.98%  13.6%      Rs 224         -16%
  8.80%   14.98%  14.7%      Rs 190         -29%

A 200 basis point rise in the 10-year G-Sec, with
NOTHING changing at the company, cuts the DCF value
by about 29%.

THE DURATION EFFECT - WHY GROWTH STOCKS FALL MORE
A company whose cash flows arrive mostly in years
1-5 is less sensitive to r than one whose cash flows
arrive in years 10-25, for exactly the same reason a
30-year bond falls more than a 2-year bond when
yields rise.

  Company X: steady Rs 100 cr FCF, flat forever
  Company Y: Rs 20 cr now, growing 25% for 10 years,
             then flat

  At r = 10%:  X = Rs 1,000 cr ;  Y = approx 1,050 cr
  At r = 13%:  X = Rs   769 cr ;  Y = approx   700 cr
  X falls 23%.  Y falls 33%.

  This is the entire mechanism behind "rate rises hurt
  growth stocks more". It is arithmetic, not sentiment.

SECOND-ORDER EFFECTS ON THE COMPANY ITSELF
  Floating-rate debt       -> interest cost rises
                              directly
  Refinancing              -> maturing debt reprices
                              upward
  Customer demand          -> housing, autos and
                              consumer durables are
                              credit-financed and fall
  Working capital cost     -> every day of receivables
                              costs more
  Competitor distress      -> leveraged competitors
```

```
weaken, which can BENEFIT
a net-cash company
```

The other macro channels

Variable	Where to track it	Transmission to a company	Who benefits / who is hurt
Inflation (CPI, WPI)	MoSPI; RBI monetary policy statements	Input costs rise before output prices can be raised, compressing gross margin in the interim. Pricing power determines the lag.	Benefits: companies with brand pricing power and inventory gains. Hurts: fixed-price contractors, regulated utilities, anyone with annual price agreements.
INR / USD	RBI reference rate	Exporters earn more rupees per dollar; importers pay more. Foreign-currency borrowers face translation losses.	Benefits: IT services, pharma exporters, textiles. Hurts: oil marketing, electronics assembly, companies with unhedged external commercial borrowings.
Crude oil	International benchmarks; Indian basket price	Directly into freight, power, packaging, and petrochemical feedstock. Also drives the current account deficit and therefore the rupee and rates.	Hurts: paints, tyres, aviation, logistics, chemicals. Benefits: upstream oil producers.
Government capex and policy	Union Budget; ministry capital outlay data; PLI scheme notifications	Order inflows for infrastructure, capital goods, cement and steel. PLI schemes shift the economics of specific manufacturing categories.	Benefits: construction, capital goods, building materials. Verify every scheme's current status at the relevant ministry — schemes are modified and withdrawn.
Monsoon and rural demand	IMD forecasts; sowing data; MSP announcements	Roughly half of India's population depends on agriculture for income. Rural demand drives two-wheelers, tractors, entry-level FMCG and gold.	A weak monsoon transmits to consumer volumes with a one-to-two quarter lag

Credit growth	RBI weekly statistical supplement; sectoral deployment data	Bank credit growth is a coincident indicator of corporate capex and consumption	Sustained credit growth well above nominal GDP growth has historically preceded asset-quality problems
Global rates and flows	FII and DII flow data from NSE and SEBI	Foreign portfolio flows affect liquidity and valuation multiples, particularly in large-caps, independently of domestic fundamentals	Domestic institutional flows from SIPs have substantially reduced this dependence over the last decade, but have not eliminated it

✦ **How to use macro without forecasting it**

Do not predict. Instead, for each company you own, write down a one-page sensitivity sheet: what happens to revenue, margin and interest cost if rates rise 200 basis points, if the rupee depreciates 10%, if the company's key input rises 30%, if demand falls 20%. Keep it with the thesis. When the macro variable actually moves, you already know what it means and you can act in hours rather than reacting to the price. **The value is in having done the work beforehand, not in having guessed the direction.**

⚠ **The honest evidence on macro forecasting**

Professional forecasts of GDP growth, inflation, interest rates and currency levels have, across many studies and many countries, shown little predictive skill beyond naive extrapolation, and they systematically fail to anticipate turning points — which is the only time a forecast would have been useful. Building a portfolio on a macro call means staking capital on the one activity in finance with the weakest documented track record. Understanding transmission is useful. Predicting the variable is not a game worth playing.

Technical Analysis Part 1 — The Honest Framing

Technical analysis is the study of price and volume history in order to say something about future price. It is taught badly, sold aggressively, and defended tribally. It also contains a small, real, academically documented core surrounded by a very large body of material that does not survive honest testing. This section separates the two, because you cannot use a tool sensibly until you know which parts of it work.

A specific warning about Indian retail derivatives

SEBI has published studies on individual traders in the equity derivatives segment showing that a large majority lose money, with aggregate retail losses running into many thousands of crores per year, and that losses grow with trading frequency. Short-horizon technical trading in futures and options is the single most reliable way for an individual to destroy capital in Indian markets. Check the latest SEBI study at **sebi.gov.in** before you form a view that you will be the exception. Nothing in this section is encouragement to trade.

The efficient market debate, stated fairly

Form of market efficiency	Claim	If true, this is useless	What the evidence actually shows
Weak form	Past prices contain no information about future prices	All of technical analysis	**Mostly supported, with documented exceptions.** Short-horizon price changes are close to unpredictable. But medium-horizon momentum (3–12 months) and long-horizon reversal (3–5 years) have been found repeatedly across markets and decades, including in Indian data.
Semi-strong form	All public information is already in the price	All of fundamental analysis	**Mostly true for large, heavily covered companies; much weaker for small, under-covered ones.** Post-earnings-announcement drift and the under-reaction to certain corporate events are well documented.
Strong form	Even private information is in the price	Insider trading	**Clearly false**, which is why insider trading is illegal and enforced against

The reasonable position

Markets are *mostly* efficient, *mostly* of the time, and least efficient where attention is scarcest — small caps, complex structures, unloved sectors, and periods of forced selling. This is not a defence of technical analysis; it is the reason fundamental work has any value at all. The same logic cuts against chart-based methods: patterns that are visible to everyone on a free charting platform are the *most* likely to have been arbitraged away, because they require no effort to find.

What has survived testing, and what has not

Technique	Evidence status	Honest assessment
Time-series momentum / trend following	**Robust.** Documented across equities, bonds, currencies and commodities, across many decades and many countries, by multiple independent research groups.	The strongest result in the whole field. Instruments trending up over the past 6–12 months have tended to continue for a few more months. **But**: it suffers long and severe drawdowns, it has weakened as it became widely known and traded, and realised returns after costs and taxes are far below backtested figures.
Cross-sectional momentum	**Robust** as a factor, including in Indian equity data	Buying past winners and selling past losers has shown a premium. Prone to violent crashes at market turning points, when the losers rally hardest.
Moving average crossovers	**Partially supported** — they are simply a lagging implementation of trend following	Work in trending markets, lose steadily in range-bound markets through repeated false signals. Useful as a trend *filter*, poor as a signal generator.
Volume confirmation	**Weakly supported.** Volume carries some information about the conviction behind a move.	A breakout on volume many times the average is more meaningful than one on thin volume. The effect is real but modest.
Support and resistance	**Weak, and partly self-fulfilling.** Round numbers and prior extremes attract orders because many participants place orders there.	Genuinely useful for one purpose: **deciding where to place a stop-loss**, since it identifies a price at which your reasoning would be demonstrably wrong. Much less useful for predicting direction.
Named candlestick patterns	**Not supported** by the systematic studies that have tested them with proper statistical controls	Vast literature, little evidence. The patterns are identified after the fact and the "success rate" figures quoted in trading courses almost never survive out-of-sample testing.
Chart patterns (head and shoulders, triangles, flags)	**Largely not supported.** Some studies find marginal effects for a few patterns; most do not replicate.	The core problem is **subjectivity**: two analysts draw different patterns on the same chart, and the pattern is only unambiguous in hindsight. A method that cannot be stated as a rule a computer could

		follow cannot be tested, and if it cannot be tested it is not evidence.
Oscillators (RSI, stochastics) as reversal signals	**Not supported.** "Overbought" instruments routinely become more overbought.	RSI is a useful *descriptive* statistic — it tells you how stretched a move has been. It is a poor *predictive* one. Buying because RSI is below 30 is a reliable way to buy things that keep falling.
Elliott Wave, Gann, Fibonacci retracements, astrology-based methods	**No credible evidence.** Unfalsifiable in practice, since the wave count or level is revised when the forecast fails.	Avoid. A method that can be re-labelled after the event to have been "correct" makes no testable claim.
Volatility measures (ATR, realised volatility)	**Supported.** Volatility clusters and is substantially more predictable than direction.	**The most genuinely useful part of the technical toolkit**, and the part least discussed. Used for position sizing and stop placement rather than for prediction.

Why most technical analysis fails: four structural reasons

1. ### Overfitting and the multiple-comparisons problem

   There are thousands of possible indicators, each with adjustable parameters, tested across thousands of instruments and timeframes. With that many combinations, some will look spectacular on historical data purely by chance. If you test 1,000 strategies at a 5% significance level, you expect 50 to appear "significant" when none of them work. Published trading systems are selected precisely because they looked good in the past, which guarantees selection bias.

2. ### Survivorship and hindsight in the examples

   Teaching material shows the charts where the pattern worked. The same pattern on the same stock three months earlier, where it failed, is not shown. Nobody keeps a log of every head-and-shoulders that did not complete.

3. ### Subjectivity defeats testing

   "Draw a trendline connecting the significant lows" is not a rule. Which lows are significant? Over what window? Where two analysts draw different lines, the method has no falsifiable content, and whatever happens can be explained afterwards.

4. ### Costs, taxes and slippage destroy the edge

   A strategy generating a 12% gross annual return with 300% turnover faces brokerage, STT, exchange charges, GST, stamp duty, bid-ask spread and short-term capital gains tax at 20% on equity (as at FY 2025-26 — verify at incometax.gov.in). The net result is frequently negative.

Any backtest that does not model full Indian transaction costs and taxes is fiction.

The defensible uses of technical tools

RISK MANAGEMENT

Defining in advance a price at which you will accept you were wrong, and sizing the position so that being wrong costs a known, survivable amount. This is the strongest case for using charts at all, and it is about discipline rather than prediction.

POSITION SIZING

Volatility-adjusted sizing — smaller positions in more volatile instruments so that each contributes similar risk. ATR is a legitimate input here. See the worked example in section 15.

EXECUTION TIMING

Once the fundamental decision to buy is made, spreading entry over weeks and avoiding buying into a vertical move is sensible. This is a marginal improvement to a decision already taken, not a decision in itself.

LIQUIDITY AND CROWDING ASSESSMENT

Average traded volume and delivery percentage tell you whether you can actually enter and exit a small-cap at a reasonable price. Entirely practical, no prediction required.

SENTIMENT CONTEXT

Extreme readings — a stock up 300% in six months on ten times normal volume — are a reason to re-examine valuation assumptions, not a signal in themselves.

A CHECK ON YOUR OWN NARRATIVE

If you believe a company is improving and the price has fallen steadily for two years on rising volume, somebody disagrees with you and may know something. That is a prompt to re-read the filings, not to capitulate.

⚠ **The strongest honest conclusion**

For an individual investor with a multi-year horizon, technical analysis is **optional**, and the opportunity cost of learning it thoroughly is high. The same hours spent reading annual reports will produce more. If you do use it, confine it to risk management and execution, keep the toolkit small, write your rules down before you trade so that they can be evaluated, and keep a log of every signal including the ones that failed. If you will not keep the log, you will never know whether it is helping, and almost everyone who does not keep the log believes it is.

Technical Analysis Part 2 — The Toolkit

What follows is a working description of the standard tools, with an honest note on the reliability of each. Learn them so that you can read what others are saying and so that you can use the two or three that genuinely help. Do not mistake familiarity with the vocabulary for an edge.

Charts and timeframes

Chart type	Shows	Best for
Line	Closing prices only	Long-term trend at a glance; least visual noise
Bar (OHLC)	Open, high, low, close	Full information without the visual bias candlesticks introduce
Candlestick	The same four values, with the open-to-close body coloured	The market standard. Easy to read; also encourages pattern-hunting.
Point and figure / Renko	Price movement only, with time removed	Filtering noise; niche

Timeframe discipline matters more than chart type. For an investor, weekly and monthly charts are the only ones worth opening — a daily chart contains mostly noise and invites action. The shorter the timeframe you watch, the more you will trade, and the more you trade the worse your net outcome is likely to be. If you hold for years, a weekly chart over ten years tells you everything a chart can tell you.

Candlestick anatomy and the common patterns

TEXT

```
CANDLESTICK ANATOMY

        |  ← upper wick (shadow): the HIGH
      _____
      |   |  ← body: open to close
      |   |     filled/red  = close below open
      _____     hollow/green= close above open
        |  ← lower wick: the LOW

  A long body    = strong directional conviction
  A short body   = indecision
  A long wick    = a move that was rejected

PATTERNS MOST OFTEN QUOTED (with honest notes)

  DOJI          open ~= close; a tiny body.
                Said to signal indecision. True
                descriptively; weak predictively.

  HAMMER        small body at the top, long lower
                wick, after a decline. Said to be
                bullish. Weak evidence.

  SHOOTING STAR mirror image after a rise.
                Weak evidence.

  ENGULFING     a body that completely covers the
                previous candle's body. The most
                commonly cited reversal pattern.
                Evidence is mixed at best; the
                effect, where found, is small and
                does not survive transaction costs.

  MARUBOZU      a long body with almost no wicks:
                the move went one way all session.
                This one is at least unambiguous
                as a description of conviction.

  MORNING /     three-candle reversal formations.
  EVENING STAR  Heavily taught; poorly evidenced.

HONEST SUMMARY: candlestick patterns describe what
happened in a session with useful compactness. Their
predictive claims are not supported by the systematic
studies that have tested them properly. Learn the
anatomy, which is genuinely informative; treat the
named patterns as vocabulary rather than as signals.
```

Support, resistance, trendlines and channels

- [ ] **Support** is a price region where buying has previously been sufficient to stop a decline; **resistance** is the mirror image. The mechanism is partly order clustering at round numbers and prior extremes, and partly the behaviour of participants who bought at a level and want to exit at break-even.
- [ ] **They are zones, not lines.** Treating ₹480 as support when the stock has turned at ₹471, ₹484 and ₹476 is false precision.
- [ ] **Broken support often acts as resistance afterwards**, and vice versa. Commonly observed, modestly evidenced.
- [ ] **Trendlines** connect successive higher lows in an uptrend or lower highs in a downtrend. The problem is that which lows you connect is a choice, and the line can usually be drawn several ways. A trendline on a **log scale** over many years is more defensible than one on a linear scale over months.
- [ ] **Channels** add a parallel line on the opposite side. Visually appealing; the same subjectivity problem.
- [ ] **The only robust use:** a clearly defined level below which your reason for owning the stock would be called into question, used to place a stop. That converts an ambiguous drawing into a concrete risk decision.

Moving averages

TEXT

```
MOVING AVERAGES

SMA(n) = arithmetic mean of the last n closes
EMA(n) = exponentially weighted, recent prices
         weighted more heavily
         multiplier k = 2 / (n + 1)
         EMA_today = (Close x k) + (EMA_yest x (1-k))

COMMONLY USED PERIODS
  20-day   about one trading month - short term
  50-day   about one quarter - intermediate trend
  100-day  intermediate
  200-day  about one year - the long-term trend line
           most widely watched by institutions

WORKED: 20-DAY SMA
  Last 20 closes sum to Rs 9,420
  SMA(20) = 9,420 / 20 = Rs 471.00
  Current price Rs 480 --> trading 1.9% above the
  20-day average: a mild short-term uptrend.

CROSSOVERS
  "Golden cross"  50-day crosses ABOVE the 200-day
  "Death cross"   50-day crosses BELOW the 200-day

  HONEST NOTE: both are LAGGING by construction - a
  50-day average cannot cross a 200-day average until
  well after the move has happened. Studies of these
  signals find results that are inconsistent across
  markets and periods, and that often disappear after
  costs. They are widely watched, which makes them
  worth KNOWING about, but they are not an edge.

A DEFENSIBLE USE: THE 200-DAY AS A REGIME FILTER
  Price above its rising 200-day average
    --> the long-term trend is up; a dip is more
        likely to be noise
  Price below its falling 200-day average
    --> the long-term trend is down; "cheap" may get
        cheaper
  This is NOT a buy or sell signal. It is context,
  and it is a reasonable prompt to re-read your
  thesis before averaging down.
```

Volume

Observation	Conventional reading	Honest note
Price rises on expanding volume	Conviction behind the move	The best-supported volume claim, and still modest
Price rises on falling volume	Weak move, prone to reversal	Weak evidence
A breakout above resistance on low volume	Likely to fail	Plausible, weakly evidenced
A large single-day volume spike	Institutional activity or news	Check the exchange's **bulk and block deal** disclosures the same day — this is actual information, not inference
Delivery percentage	The share of traded volume taken to demat rather than squared off intraday	**India-specific and genuinely useful.** Published by NSE and BSE daily. A high delivery percentage means real ownership transfer rather than intraday churn. Persistently very low delivery percentages in a small-cap with a rising price is a warning sign.
Average daily traded value	Liquidity	Practical and important. If a stock trades ₹2 crore a day and you want a ₹20 lakh position, exiting in a panic will cost you several percent. Size positions against liquidity.

Indicators: RSI, MACD, Bollinger Bands, ATR

TEXT

```
RSI - RELATIVE STRENGTH INDEX (14-period default)

  RS  = average gain over n periods
        / average loss over n periods
  RSI = 100 - (100 / (1 + RS))

  WORKED (simplified 14-day)
    Sum of up-day moves    = Rs 74
    Sum of down-day moves  = Rs 38
    Average gain = 74/14 = 5.29
    Average loss = 38/14 = 2.71
    RS  = 5.29 / 2.71 = 1.95
    RSI = 100 - (100 / 2.95) = 100 - 33.9 = 66.1

  Conventional reading: above 70 "overbought",
  below 30 "oversold".

  HONEST NOTE: this is the most misused indicator in
  the field. In a strong trend, RSI sits above 70 for
  MONTHS while the price keeps rising, and below 30
  for months while it keeps falling. Selling at RSI 70
  in a structural uptrend is a reliable way to exit
  the best positions you will ever own. RSI describes
  how stretched a move has been. It does not predict
  that it will stop.

MACD - MOVING AVERAGE CONVERGENCE DIVERGENCE

  MACD line   = EMA(12) - EMA(26)
  Signal line = EMA(9) of the MACD line
  Histogram   = MACD line - signal line

  WORKED
    EMA(12) = Rs 476.40
    EMA(26) = Rs 468.90
    MACD    = 7.50
    Signal (EMA9 of MACD) = 6.20
    Histogram = +1.30  -> short-term momentum is
                          improving

  HONEST NOTE: MACD is two moving averages subtracted
  from each other. It is a trend-following tool
  wearing a disguise, and it inherits every weakness
  of moving averages, including whipsaws in
  range-bound markets.

BOLLINGER BANDS
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```
  Middle = SMA(20)
  Upper  = SMA(20) + 2 standard deviations
  Lower  = SMA(20) - 2 standard deviations

  WORKED
    SMA(20)            = Rs 471.00
    20-day std dev     = Rs  18.40
    Upper band         = 471 + 36.80 = Rs 507.80
    Lower band         = 471 - 36.80 = Rs 434.20
    Band width as % of middle
                       = 73.60 / 471 = 15.6%

  USEFUL READING: band width measures VOLATILITY. A
  contraction ("squeeze") reliably indicates that
  volatility is low, and volatility clusters - so a
  low-volatility period is more likely to be followed
  by a higher-volatility one. That is a real
  statistical property.

  NOT A USEFUL READING: "price touched the upper band,
  so sell." In a trend, price walks along the upper
  band for weeks. The bands say nothing about
  direction.

ATR - AVERAGE TRUE RANGE (the genuinely useful one)

  True Range = the greatest of:
      (high - low)
      |high - previous close|
      |low  - previous close|
  ATR(14) = 14-period average of True Range

  WORKED
    ATR(14) = Rs 14.20 on a price of Rs 480
    ATR as % of price = 2.96%
    → this stock moves about 3% on a typical day

  POSITION SIZING WITH ATR - THE REAL APPLICATION
    Portfolio value                  Rs 20,00,000
    Maximum risk per position (1%)   Rs    20,000
    Stop placed at 2.5 x ATR below entry
      = 2.5 x 14.20 = Rs 35.50 below Rs 480
      = stop at Rs 444.50
    Risk per share = Rs 35.50
    Position size  = 20,000 / 35.50 = 563 shares
    Capital deployed = 563 x 480 = Rs 2,70,240
                     = 13.5% of the portfolio

    CONTRAST - a low-volatility stock
```

```
  Price Rs 480, ATR Rs 6.00 (1.25% of price)
  Stop at 2.5 x 6.00 = Rs 15.00 below
  Risk per share = Rs 15.00
  Position size  = 20,000 / 15 = 1,333 shares
  Capital deployed = Rs 6,39,840 = 32% of portfolio

THE POINT: both positions risk exactly Rs 20,000 if
the stop is hit, despite one being more than twice
the size of the other. This equalises RISK rather
than equalising RUPEES, and it is the single most
defensible use of a technical indicator anywhere in
this guide. Note the practical limits: a 32%
position violates concentration limits for most
investors, gaps can take you through a stop, and
stop-losses on illiquid Indian small-caps often
execute far below the trigger price.
```

Chart patterns and gaps, catalogued with caveats

Pattern	Conventional description	Honest reliability note
Head and shoulders	Three peaks, the middle highest, with a "neckline"; a break below the neckline is said to be a reversal signal	The most famous pattern in the field. Systematic testing finds results that are weak, inconsistent, and highly sensitive to how the pattern is defined. Because identification is subjective, the pattern is reliably visible only after the move.
Double top / double bottom	Two failures at roughly the same level	Describes a real phenomenon — a level where supply or demand appeared twice — but the predictive claim is weak
Flags and pennants	A brief consolidation within a strong move, said to continue in the original direction	Consistent with momentum, which is itself documented. This is probably the most defensible of the named patterns, and even then the effect is attributable to momentum rather than to the shape.
Triangles (ascending, descending, symmetrical)	Converging price range before a breakout	A converging range is simply falling volatility, which does cluster. The direction of the eventual breakout is close to a coin flip; the claim that an ascending triangle breaks upward is not well supported.
Cup and handle	A rounded base followed by a shallow pullback	Popularised in growth-investing literature. Evidence is anecdotal and the definition is elastic.
Rounding top / bottom	Gradual change of trend	Descriptive. No testable rule.
Gaps	A price range with no trading, usually from overnight news. Classified as breakaway, runaway and exhaustion.	The classification is entirely retrospective — you only know which type it was afterwards. The useful fact is simpler: **gaps are a risk-management problem**, because they can take you straight through a stop-loss. In India, gaps around results, block deals and circuit-limit changes are common.

✦ **If you use any of this, keep a trade log**

Record, before each decision: the signal, the entry price, the stop, the target, the position size, and the reasoning in one sentence. Afterwards record the exit and the outcome. After fifty entries you will have something almost nobody has — evidence about whether your method works. Most people who believe technical analysis works for them have never kept this record, and memory is systematically biased towards the trades that worked. **The log is the whole experiment.** Add a column for total costs (brokerage, STT, exchange charges, GST, stamp duty, slippage) and for tax, because a method that is gross-positive and net-negative is worse than doing nothing.

Putting It Together: The Investment Thesis

Analysis that is not written down is not analysis; it is a feeling with spreadsheets attached. The discipline of compressing everything into one page forces you to identify what you are actually betting on, and the dated record is the only mechanism by which you will ever learn whether your judgement is any good.

The one-page stock note

TEXT

STOCK NOTE — BHARAT WIDGETS LTD
Date: 21 October 2026 | Price: Rs 480
Market cap: Rs 5,760 cr | Author: [you]

1. WHAT THE BUSINESS DOES (two sentences, no jargon)
   Manufactures precision engineering components for
   industrial OEMs across four plants in western and
   southern India. Roughly 74% domestic, 26% export.

2. WHY IT MAKES MONEY (the moat, in one sentence)
   Components are designed into customer platforms
   and re-qualification costs the customer 9–14
   months, which produces switching costs visible
   as a stable 52–53% gross margin across a decade
   including the FY2020 downturn.

3. THE NUMBERS THAT MATTER
   Revenue CAGR 9yr   10.1%
   EBITDA margin, FY16 → FY25   16.8% → 21.0%
   ROCE FY2025   18.4%
   Post-tax ROIC vs WACC   13.8% vs 12.5%
   Incremental ROCE FY21–25 post-tax   26.2%
   Net debt / EBITDA   1.13x
   Cumulative CFO / cumulative PAT, 10yr   1.41x
   Promoter holding / pledge   58.4% / nil
   Sustainable growth rate (ROE x retn)   12.75%

4. WHAT I AM BETTING ON (be specific)
   That switching costs hold gross margin above 50%
   while the new plant lifts capacity 35% by FY2028,
   and that incremental capital continues to earn
   above 20% post-tax.

5. WHAT THE PRICE IMPLIES (reverse DCF)
   At Rs 480 the market requires 16.5% revenue CAGR
   for ten years, EBIT margin reaching 20% (matching
   the best operator in the peer set), AND capex
   falling to 6.5% of revenue. Those three
   assumptions are in tension with one another.

6. VALUATION
   DCF base case   Rs 268 / share
   Plausible DCF band   Rs 204 – 372
   Probability-weighted   Rs 276
   Peer median P/E   19.8x vs BWL 24.2x
   CONCLUSION: a good business at a demanding price.
   NOT A BUY AT Rs 480. Revisit below Rs 300.

```
7. FALSIFICATION – WHAT WOULD PROVE ME WRONG
   a) Gross margin below 50% for two consecutive
      quarters without a stated one-off cause
   b) Receivable days above 75 (currently 63)
   c) CFO / EBITDA below 60% for a full year
   d) Any promoter pledge appearing
   e) New plant capex exceeding Rs 450 cr (guided
      Rs 380 cr) or slipping beyond FY2029
   f) Incremental ROCE on the expansion below 15%
      pre-tax once the plant stabilises
   g) A second acquisition at above 10x EV/EBITDA

8. PRE-MORTEM
   "It is 2030 and this investment lost 60%. What
   happened?"  Most likely: a Chinese or Vietnamese
   competitor undercut on price in the two largest
   product lines, OEMs used that as leverage in
   annual negotiations, gross margin fell to 44%,
   the new plant ran at 55% utilisation, and the
   stock de-rated from 24x to 11x on earnings that
   had themselves halved. Second most likely: an
   acquisition funded with debt at a high multiple.

9. POSITION SIZE IF BOUGHT
   Conviction: medium (business high, price poor)
   Target weight: 0% at Rs 480.
   At Rs 290 with the thesis intact: 4% of
   portfolio, built over three tranches.

10. REVIEW TRIGGERS
   Every quarterly result; the FY2027 annual report;
   any announcement touching items 7(a) to 7(g).
```

Pre-mortem and falsification

Two disciplines do more to improve outcomes than any amount of additional model-building, and both take fifteen minutes.

THE PRE-MORTEM

Before buying, assume it is five years later and the position has lost 60%. Write the story of how. This defeats the natural tendency to build a case for a conclusion you have already reached, because it *requires* you to construct the opposing argument. If you cannot write a plausible failure story, you do not understand the business well enough to own it.

FALSIFICATION CRITERIA

Specific, observable, dated conditions which, if met, mean the thesis is broken. "The stock falls 30%" is not a falsification criterion — price is not evidence about the business. "Gross margin falls below 50% for two consecutive quarters" is, because it is the measurable manifestation of the thing you claimed was durable.

Position sizing from conviction and risk

Conviction level	What justifies it	Typical weight	Maximum
High	Business understood in depth, ten-year record clean, moat identifiable and tested through a downturn, governance clean, bought at a meaningful discount to a conservative valuation	6–10%	12% at cost
Medium	Good business, fair price, one or two unresolved questions	3–5%	6%
Starter / learning	Interesting, incompletely researched; the position exists to force attention	1–2%	2%
Any governance concern	—	**0%**	**0%**
Low-liquidity small-cap	Size against daily traded value, not against conviction	Not more than one to two days of average traded value, so exit is possible	—

⚠ **Concentration and diversification, stated plainly**

A concentrated portfolio of eight to twelve well-understood businesses is defensible for someone who genuinely does thirty hours of work per company per year. A concentrated portfolio held by someone who does not is simply leveraged exposure to their own blind spots. **Most people overestimate how much work they do.** The honest default for an individual with a full-time job is a core of index or diversified funds (see the funds guide) with a deliberately capped satellite sleeve — perhaps 10–25% of the equity allocation — for direct stocks. That cap is also what makes the direct sleeve a useful learning exercise rather than a threat to your financial plan.

What would make you sell

Reason	Legitimate?	Note
A falsification criterion was met	**Yes**	This is what they are for. Act without renegotiating with yourself.
The thesis played out and the price now exceeds any plausible valuation	**Yes**	Re-run the reverse DCF. If the implied assumptions have become absurd, the work is done.
A materially better opportunity exists and capital is fully deployed	**Yes**	Opportunity cost is real. Account for the tax on the switch.
Governance deteriorated: pledge, auditor resignation, related-party escalation	**Yes, immediately**	Do not wait for confirmation. The asymmetry is extreme.
Position size has grown beyond your risk limit	**Yes, trim**	Trimming is not the same as selling
The price fell	**No**	Price is not information about the business. Re-check the thesis; if intact, a lower price is better.
The price rose and you want to "book profits"	**No**	The most expensive habit in retail investing: selling winners early and holding losers. Round-tripping also triggers tax and resets your compounding.
The stock has gone nowhere for two years	**No**	Two flat years is normal and is precisely the period during which your structural advantage of a long horizon is being exercised
Someone on television or a messaging group said to	**No**	—

Screening

A screen is a hypothesis expressed as filters. It narrows thousands of listed companies to a few dozen worth reading about. It is a starting point and never an answer, because

every screen operates on standardised data that is sometimes wrong, and every popular screen is run by thousands of people simultaneously.

Build the screen from a hypothesis

TEXT

```
WRONG WAY: open a screener, tick some boxes, see
what appears.

RIGHT WAY: state the hypothesis first, then express
it as filters, then ask what the filters will let
through that you do not want.

HYPOTHESIS: "Mid-sized Indian manufacturers that
have improved returns on capital over five years
while deleveraging are under-appreciated, because
screens and headlines focus on growth rather than
on capital efficiency."

FILTERS
  Market cap                 Rs 1,000 - 20,000 cr
  ROCE (latest)              > 15%
  ROCE (5 years ago)         < ROCE (latest)
  Net debt / EBITDA          < 2.0x
  Net debt / EBITDA 5y ago   > net debt / EBITDA now
  Cumulative CFO / cumulative PAT, 5 years  > 1.0
  Promoter pledge            = 0
  Revenue CAGR 5 years       > 6%
  Sales                      > Rs 500 cr

WHAT THIS SCREEN WILL STILL LET THROUGH - and you
must check each manually:
  - Cyclicals at the top of their cycle, where ROCE
    is temporarily high and will revert
  - Companies whose ROCE rose because they sold
    assets rather than because operations improved
  - Companies that deleveraged by issuing equity,
    diluting you
  - Data errors, especially in companies with
    unusual structures or recent demergers
  - Businesses in structural decline where the
    improvement is cost-cutting with no runway
  - Related-party-heavy companies: no screener
    field captures this

OUTPUT: typically 20-60 names. Read the one-page
summary for each. Discard most within five minutes
on the three cash cross-checks from section 11.
Fully research three to five.
```

The common screen archetypes and their failure modes

Screen	Typical filters	What it finds	Failure mode
Quality	ROCE above 20% for ten years, debt-to-equity under 0.3, CFO/PAT above 1, consistent margins	Genuinely good businesses	They are almost always expensive. The screen finds the right companies at the wrong price, and quality at 60× earnings has produced poor returns many times.
Value	P/E under 12, P/B under 1.5, dividend yield above 3%	Statistically cheap companies	**Value traps.** A large share of what this screen returns is cheap because the business is deteriorating. Pair with a quality or improvement filter and with a cash-conversion check.
Growth	Revenue and PAT CAGR above 20%, margin expanding	Fast-growing companies	Catches acquisitive roll-ups, companies recovering from a low base, and cash-burning growth. Always strip out inorganic growth and check cumulative FCF.
Turnaround	Losses in prior years, latest quarters profitable, falling debt	Possible inflections	Most turnarounds do not turn. Highest failure rate of any archetype; requires the deepest research.
Dividend	Yield above 4%, payout under 60%, ten years of uninterrupted dividends	Mature cash generators	A high yield is frequently a forecast of a cut. Check that the dividend is covered by free cash flow, not by borrowing.
Magic-formula style	Rank on earnings yield and on return on capital, combine the ranks	A mechanical value-plus-quality list	Works in backtests, suffers long stretches of underperformance, and is heavily affected by data errors and by financial companies that must be excluded

Momentum	Top decile of 12-month price return excluding the most recent month	Trending stocks	High turnover, high transaction costs and short-term capital gains tax in India, plus periodic violent crashes at turning points

✕ **Three rules for screening in India**

One: exclude banks, NBFCs and insurers from any general screen — their ratios are not comparable and will dominate or distort the output. **Two: verify every number on the shortlist against the annual report** before spending real research time; standardisation errors in Indian screener data are common, especially around exceptional items, consolidated-versus-standalone, and companies with recent corporate actions. **Three: a screen cannot see governance.** Related-party transactions, pledge trends, auditor changes and promoter history have no screener field, and they are the factors most likely to cause a total loss. The screen's job is to produce a reading list, nothing more.

Frequently Asked Questions

Questions a beginner analysing Indian stocks actually asks, answered without hedging. None of this is a recommendation; all rates and rules are as at FY 2025-26 / late 2026 and must be verified at the official portals named throughout.

How many years of financials do I really need to look at?

Ten, as a hard minimum, and more if the company is cyclical. Five years will usually not include a downturn, and a business that has never been tested has not told you anything about its durability. The specific thing you are looking for in the long series is **behaviour in the worst year**: what happened to volumes, margins, receivables, debt and the dividend. That single year tells you more about the business than the nine good ones combined.

Can I rely on screener websites instead of reading annual reports?

For generating a reading list, yes. For making a decision, no. Screeners standardise Indian line items into their own schema and routinely mis-handle exceptional items, other income, consolidated-versus-standalone, lease liabilities and companies with recent demergers or acquisitions. More importantly, **no screener has a field for related-party transactions, pledge trends, auditor resignations, contingent liabilities or the CWIP ageing schedule** — and those are the items most likely to cause you a permanent loss. Use the screener for step one, the annual report for every step after.

Should I use standalone or consolidated numbers?

Consolidated, always, as your primary view — it describes the whole economic entity you own a share of. Use standalone only as a diagnostic to isolate how the parent's own operations are doing versus the subsidiaries. A company whose standalone accounts look healthy while consolidated accounts show losses has problems in subsidiaries, which is a pattern worth investigating rather than ignoring. Note that many screeners default to standalone.

What is the single most useful ratio if I only have time for one?

Cumulative cash flow from operations divided by cumulative profit after tax, over ten years. It should be comfortably above 1.0, because depreciation is added back. If it is below 1.0 over a decade, reported profits are not becoming cash, and nothing else in the analysis matters until you understand why. It takes about four minutes per company and it removes a large share of the worst outcomes from your opportunity set. If you have time for a second, make it post-tax ROIC versus WACC.

Is a low P/E a buying signal?

Almost never on its own. Work through the seven structural causes of a low multiple in section 8: low return on equity, high risk, low growth, cyclically peak earnings, governance doubt, illiquidity, and poor earnings quality. In the overwhelming majority of cases at least one applies, and the market is pricing it correctly. For cyclical businesses the relationship actually inverts — a steel or cement company at 6× trailing earnings is usually at the top of its cycle, and a low P/E there is a warning rather than an invitation.

How accurate is a DCF really?

As a point estimate, close to useless — the worked example in section 9 produces a range of ₹204 to ₹372 across nothing more than plausible variations in the discount rate and terminal growth, before touching any operating assumption. Its genuine value is in two places. First, it forces every assumption into the open where it can be argued about. Second, run in reverse, it converts the market price into a specific testable statement about growth and margins, which is a question you can actually research. Build the DCF for the discipline; use the reverse DCF for the decision.

What discount rate should I use for an Indian company?

For a typical listed non-financial company, the plausible band is roughly 11–14%, built from a 10-year G-Sec yield (verify the current level at rbi.org.in) plus an equity risk premium for India that reasonable practitioners put somewhere between about 5.5% and 8%, adjusted for the company's beta and capital structure. Add a further premium for small size, illiquidity or governance concerns if applicable. The exact number is less important than **testing the valuation across the band and reporting the range**. Anyone quoting a precise WACC to two decimal places is conveying false confidence.

Does technical analysis work?

A narrow part of it does. Time-series and cross-sectional momentum are robustly documented across many markets and decades, and volatility clustering is a real statistical property that supports position sizing with measures like ATR. Most of the rest — named candlestick patterns, head and shoulders, oscillator reversal signals, Elliott Wave, Gann — does not survive systematic testing, largely because it

is too subjective to test and the teaching material shows only the cases that worked. For an investor with a multi-year horizon, the defensible uses are risk management, position sizing and execution timing. **If you use it, keep a log including the failed signals, because memory is systematically biased towards the ones that worked.**

Why are bank ratios so different?

Because a bank's raw material is money and its biggest cost is a judgement about future loan losses. Deposits are not distress debt — leverage of 8–12× is the business model, so debt-to-equity is meaningless. Interest income is operating revenue, so EBITDA has no meaning. The maturity mismatch between short deposits and long loans is the business, so the current ratio is meaningless. Use instead: net interest margin, CASA ratio, GNPA and NNPA, provision coverage, credit cost, capital adequacy and CET-1, cost-to-income, return on assets, and price to adjusted book value. Section 12 has the full set.

What are the fastest red flags to check before spending real time on a company?

Six checks, roughly ten minutes total. (1) Cumulative CFO versus cumulative PAT over ten years. (2) The promoter pledge column in the latest shareholding pattern. (3) The auditor's report for a qualification, an emphasis of matter on going concern, or a recent resignation. (4) The related-party transactions note, as a percentage of revenue and purchases, trended. (5) Receivable and inventory days over ten years against revenue growth. (6) The contingent liabilities note against equity, looking specifically for guarantees given for promoter group entities. Any one of these failing is sufficient reason to stop and move to the next idea.

How do I value a loss-making company?

Very carefully, and usually you should not try. The honest options are: a DCF that explicitly models the path to profitability with the funding requirement included, which is extremely assumption-dependent; EV/Sales benchmarked against what mature peers achieve, which requires you to assume a terminal margin; unit economics built bottom-up (contribution per customer, customer acquisition cost, payback period, retention), which is the most informative when the data is disclosed; or replacement cost and asset value for a cyclical at the trough. In every case, model the **cash runway and the dilution required to reach breakeven**, because a company that must raise equity at a depressed price transfers a large share of the eventual upside away from existing holders.

How much of my portfolio should be in direct stocks?

This is a question about your time, not your intelligence. A first pass on one company takes 26–42 hours, and an annual update takes about five. Fifteen companies is therefore a part-time job indefinitely. For most working professionals the honest answer is to hold the core of the equity allocation in diversified or index funds and cap the direct-stock sleeve at something like 10–25% of equity — enough to learn properly and to care about the outcome, small enough that your blind spots cannot damage the financial plan. The mutual funds guide covers the core allocation in detail, and the foundations guide covers the question of how much to put into equity at all.

What should I do when a stock I own falls 40%?

First, separate the two cases, because they require opposite responses. If the business thesis is intact and nothing on your falsification list has triggered, a 40% fall is a lower price for the same cash flows and is an opportunity — provided the position size still fits your limits. If a falsification criterion *has* triggered, the fall is information and you should act on the criterion, not on the price. The disciplined procedure is to re-read your original note before looking at any news, decide whether anything on the list has changed, and only then look at the price. **Never average down on a position where leverage or governance is part of the concern**, because those are the situations where a 40% loss becomes a 95% loss.

Are IPOs worth analysing?

The DRHP is one of the most information-dense documents available and is worth reading even if you never subscribe — particularly for the industry section and the risk factors, and particularly when the issuer is a competitor of a company you already follow. As an investment, bear in mind the structural asymmetry: the issue is timed and priced by selling shareholders and merchant bankers with far better information than you, and typically in a favourable market. There is also no long public track record to analyse, which removes most of the tools in this guide. None of that makes IPOs uninvestable; it means the burden of proof is higher, not lower, than for a company with ten years of audited filings.

How do I know when my analysis is good enough to act on?

Three tests. **One:** can you explain the business, its moat and its main risk to someone unfamiliar with it in three minutes, without notes? **Two:** can you write the pre-mortem — a specific, plausible story of how this loses 60% — without reaching? **Three:** have you written down falsification criteria that are observable and dated, so that in two years you will be able to tell whether you were right for the right reasons? If any of the three fails, you have a feeling rather than a thesis. The written note is not administrative overhead; it is the only instrument that will ever tell you whether your judgement is improving.

The Complete Analysis Checklist and a Six-Week Programme

Everything in this guide, compressed into a working checklist, followed by a structured programme for building the skill on one real company. Work through the checklist in order; most companies will fail at section A or B, which is the point — the checklist exists to save you time, not to be completed.

A. Information gathered (5 items)

- [ ] Ten years of annual reports downloaded, consolidated
- [ ] Last four concall transcripts read, with guidance numbers extracted and dated
- [ ] Latest investor presentation read, operating metrics noted, abandoned metrics noted
- [ ] Latest credit rating rationale read
- [ ] Last eight quarterly shareholding patterns checked for promoter holding and pledge trend

B. Fast disqualifiers — stop if any fails (8 items)

- [ ] Cumulative CFO / cumulative PAT over ten years is above 1.0
- [ ] Promoter pledge is zero, or small, declining and explained
- [ ] Auditor's opinion is unmodified; no resignation; no going-concern emphasis of matter
- [ ] Related-party transactions are small, at arm's length, and not rising as a share of revenue
- [ ] No loans, advances or guarantees to promoter group entities
- [ ] Contingent liabilities are modest relative to equity, with no large group guarantees
- [ ] Receivable days and inventory days are not compounding faster than revenue
- [ ] No exchange surveillance action (ASM, GSM) in force

C. Profit and loss (7 items)

- [ ] Ten-year revenue, gross, EBITDA, EBIT and PAT margin table built by hand
- [ ] Gross margin trend examined against revenue growth (the two-line test)
- [ ] Other income quantified as a percentage of PBT for each of ten years
- [ ] Exceptional items counted over ten years and normalised earnings computed
- [ ] Effective tax rate compared with the statutory rate; the reconciliation note read
- [ ] Diluted share count tracked over ten years; the dilution rate computed

- [ ] "Other expenses" note read and decomposed; "miscellaneous" checked for size and growth

D. Balance sheet (8 items)

- [ ] Net debt computed including lease liabilities; maturity ladder built from the borrowings note
- [ ] Goodwill and intangibles as a percentage of equity; impairment assumptions read
- [ ] Internally generated intangibles and "intangibles under development" examined
- [ ] CWIP ageing schedule and the overdue/over-budget project disclosure read
- [ ] Trade receivable ageing read; provision as a percentage of receivables trended
- [ ] Net operating working capital and incremental NWC per rupee of incremental revenue computed
- [ ] Funding match checked: equity plus non-current liabilities against non-current assets
- [ ] Equity movement reconciled to PAT less dividends less buybacks less OCI

E. Cash flow (5 items)

- [ ] Ten-year CFO, capex, FCF table built; the three-section signature pattern identified
- [ ] CFO/EBITDA and CFO/PAT computed for each of ten years
- [ ] Capex to depreciation ratio checked against actual capacity additions
- [ ] Maintenance versus growth capex estimated; owner earnings computed
- [ ] Dividend tested against FCFE for affordability

F. Ratios and returns (7 items)

- [ ] ROE, ROCE and post-tax ROIC computed on average balances, with definitions written down
- [ ] Post-tax ROIC compared against estimated WACC; the spread stated
- [ ] Five-step DuPont decomposition over five years; the source of any ROE change identified
- [ ] Cash conversion cycle computed and trended; compared against direct peers only

- [ ] Net debt/EBITDA, interest coverage and debt service coverage computed
- [ ] Incremental ROCE over five years computed
- [ ] Sustainable growth rate (ROE × retention) computed and compared with actual growth

G. Growth quality (4 items)

- [ ] Organic growth separated from acquired growth using the business combinations note
- [ ] Volume growth separated from price and mix, using disclosed operating metrics
- [ ] Degree of operating leverage computed; a 20–25% revenue decline stress-tested through to interest coverage
- [ ] Any past acquisition assessed for return on the price paid, and for subsequent impairment

H. Qualitative (6 items)

- [ ] The moat named specifically, with the financial signature that evidences it
- [ ] Ten-year ROCE variance examined — persistence, not level, is the moat test
- [ ] Five forces assessed, with customer and supplier concentration quantified
- [ ] TAM built bottom-up; implied market share at the forecast growth rate computed and sanity-checked
- [ ] Ten-year capital allocation scorecard built: sources, uses, and the five questions scored
- [ ] Board, auditor, remuneration and promoter conduct history reviewed

I. Valuation (6 items)

- [ ] Peer table built with every multiple computed identically by you
- [ ] Trailing and forward P/E, EV/EBITDA, EV/EBIT and FCF yield computed
- [ ] Mid-cycle earnings used instead of trailing earnings if the business is cyclical
- [ ] DCF built with every assumption written down *before* the output was computed

- [ ] Sensitivity table produced across WACC and terminal growth; both terminal-value methods reconciled
- [ ] **Reverse DCF run; the implied growth and margin assumptions stated in one sentence and judged**

J. Decision and discipline (5 items)

- [ ] One-page note written and dated
- [ ] Pre-mortem written: a specific story of a 60% loss
- [ ] Falsification criteria listed, observable and measurable
- [ ] Position size set from conviction, liquidity and volatility — not from enthusiasm
- [ ] Review triggers diarised: results dates and the specific announcements that would matter

A six-week practice programme

1. **Week 1 — Choose and understand**
   Pick one mid-sized, non-financial, profitable Indian company whose product you can describe in a sentence. Avoid banks, NBFCs, holding companies and anything loss-making for your first attempt. Read the two most recent annual reports in the order given in section 2. Write a 300-word description of what the business does and how it makes money. Do not look at the share price all week.

2. **Week 2 — Build the data sheet**
   Download ten years of annual reports. Type — do not import — the raw P&L, balance sheet and cash flow lines into a spreadsheet. Compute every derived ratio yourself. Where your number disagrees with a screener's, find out which is right and why. This week is tedious and it is where the skill actually forms.

3. **Week 3 — Interrogate the quality**
   Run every item in checklist sections B, C, D and E. Read the related-party note, the contingent liabilities note, the CWIP and receivable ageing schedules, and the auditor's Key Audit Matters for each of the last five years. Write a one-page memo listing every unresolved question.

4. **Week 4 — Industry and moat**

Read the last four concalls of your company and of three competitors. Find independent industry volume data from an association or regulator. Build the peer comparison table. Name the moat, state the financial evidence for it, and write the three specific ways it could erode.

5

Week 5 — Valuation

Build the peer multiples table, then the DCF with every assumption written down first, then the sensitivity table, then the three scenarios. Finally run the reverse DCF and write one sentence describing what the current market price requires the company to achieve. Compare that sentence against what you learned in weeks 2 to 4.

6

Week 6 — Write it up, and do nothing

Produce the one-page note from section 16, including the pre-mortem and the falsification criteria. Then

do not buy anything.

File the note. Diarise a review for each of the next four quarterly results. The purpose of the first company is to build the process, not to take a position — and a note written without the pressure of an open position is the most honest one you will ever write. Repeat the whole cycle on a second company; by the third, it takes a quarter of the time.

10	61	6	1
YEARS OF DATA	CHECKLIST ITEMS	WEEKS PER COMPANY	PAGE NOTE

✓ **The honest summary of this entire guide**

Analysis will not make you a good investor on its own. It removes the worst outcomes — the frauds, the leveraged cyclicals at the peak, the cash-burning growth stories, the companies whose profits never became cash — and that is worth a great deal, because avoiding permanent losses matters more to a long-run compounding record than catching winners. What it cannot do is tell you the future. Hold the conclusions loosely, write down why you believed them, revisit the record honestly, and keep the position sizes small enough that being wrong is survivable. And if after a year of this work you find you do not enjoy it and do not have the hours, that is a legitimate and common conclusion: the funds route is not a consolation prize, it is the correct answer for most people.

Final reminder

This guide is financial education and is not investment advice. No security, sector or strategy described here is recommended. All worked examples use invented companies. Past performance does not predict future returns and you can lose your entire capital. Every rate, tax, regulation and threshold mentioned is as at FY 2025-26 / late 2026 and must be verified at **sebi.gov.in**, **nseindia.com**, **bseindia.com**, **rbi.org.in**, **mca.gov.in** and **incometax.gov.in** before you rely on it. For advice specific to your circumstances, consult a **SEBI-registered Investment Adviser** and verify their registration number before paying anything.